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LIBRARY: sky130_fd_sc_hd__tt_025C_1v80.lib

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Warning 1171:
/data/Desktop/work/tools/openlane_working_dir/OpenSTA/vsdbabysoc/lib/sky130_fd_sc_hd__tt_0
25C_1v80.lib line 23, default_fanout_load is 0.0.

Warning 1140:
/data/Desktop/work/tools/openlane_working_dir/OpenSTA/vsdbabysoc/lib/sky130_fd_sc_hd__tt_0
25C_1v80.lib line 1, library sky130_fd_sc_hd__tt_025C_1v80 already exists.

Warning 1171:
/data/Desktop/work/tools/openlane_working_dir/OpenSTA/vsdbabysoc/lib/sky130_fd_sc_hd__tt_0
25C_1v80.lib line 23, default_fanout_load is 0.0.

Warning 1140:
/data/Desktop/work/tools/openlane_working_dir/OpenSTA/vsdbabysoc/lib/avsddac.lib line 1, library
avsddac already exists.

Warning 1140:
/data/Desktop/work/tools/openlane_working_dir/OpenSTA/vsdbabysoc/lib/avsdpll.lib line 1, library
avsdpll already exists.

----- REPORT CHECKS -----

Warning 168: run_final.tcl line 1, unknown field nets.

Startpoint: core/CPU_imm_a2_reg[10]

(rising edge-triggered flip-flop clocked by clk)

Endpoint: core/CPU_imm_a3_reg[10]

(rising edge-triggered flip-flop clocked by clk)

Path Group: clk

Path Type: min

Fanout	Cap	Slew	Delay	Time	Description
--------	-----	------	-------	------	-------------

	0.00	0.00	0.00		clock clk (rise edge)
		0.00	0.00		clock network delay (ideal)
	0.00	0.00	0.00		^ core/CPU_imm_a2_reg[10]/CLK (sky130_fd_sc_hd__dfxtp_1)
1	0.00	0.03	0.27	0.27	^ core/CPU_imm_a2_reg[10]/Q (sky130_fd_sc_hd__dfxtp_1)
	0.03	0.00	0.27		^ core/CPU_imm_a3_reg[10]/D (sky130_fd_sc_hd__dfxtp_1)
			0.27		data arrival time
	0.00	0.00	0.00		clock clk (rise edge)
		0.00	0.00		clock network delay (ideal)
	0.00	0.00			clock reconvergence pessimism
		0.00			^ core/CPU_imm_a3_reg[10]/CLK (sky130_fd_sc_hd__dfxtp_1)
	-0.03	-0.03			library hold time
		-0.03			data required time

		-0.03			data required time
		-0.27			data arrival time

		0.31			slack (MET)

Startpoint: core/CPU_is_addi_a3_reg

(rising edge-triggered flip-flop clocked by clk)

Endpoint: core/CPU_Xreg_value_a4_reg[27][30]

(rising edge-triggered flip-flop clocked by clk)

Path Group: clk

Path Type: max

Fanout	Cap	Slew	Delay	Time	Description
--------	-----	------	-------	------	-------------

	0.00	0.00	0.00		clock clk (rise edge)
	0.00	0.00			clock network delay (ideal)
	0.00	0.00	0.00		^ core/CPU_is_addi_a3_reg/CLK (sky130_fd_sc_hd__dfxtp_1)
2	0.01	0.04	0.28	0.28	v core/CPU_is_addi_a3_reg/Q (sky130_fd_sc_hd__dfxtp_1)
	0.04	0.00	0.28		v core/U474/A (sky130_fd_sc_hd__nor2_1)
2	0.00	0.11	0.12	0.40	^ core/U474/Y (sky130_fd_sc_hd__nor2_1)
	0.11	0.00	0.40		^ core/U476/A (sky130_fd_sc_hd__nand2_1)
2	0.01	0.08	0.10	0.50	v core/U476/Y (sky130_fd_sc_hd__nand2_1)
	0.08	0.00	0.50		v core/U9/A (sky130_fd_sc_hd__inv_2)
34	0.15	0.67	0.51	1.01	^ core/U9/Y (sky130_fd_sc_hd__inv_2)
	0.67	0.00	1.01		^ core/U479/B (sky130_fd_sc_hd__xor2_1)
2	0.01	0.17	0.18	1.19	v core/U479/X (sky130_fd_sc_hd__xor2_1)
	0.17	0.00	1.19		v core/U569/A2 (sky130_fd_sc_hd__a21oi_1)
2	0.01	0.19	0.25	1.43	^ core/U569/Y (sky130_fd_sc_hd__a21oi_1)
	0.19	0.00	1.43		^ core/U571/A2 (sky130_fd_sc_hd__o21ai_1)
2	0.01	0.08	0.11	1.54	v core/U571/Y (sky130_fd_sc_hd__o21ai_1)
	0.08	0.00	1.54		v core/U574/A1 (sky130_fd_sc_hd__a21oi_1)
2	0.01	0.19	0.19	1.74	^ core/U574/Y (sky130_fd_sc_hd__a21oi_1)

0.19 0.00 1.74 ^ core/U576/A2 (sky130_fd_sc_hd__o21ai_1)

2 0.01 0.08 0.11 1.84 v core/U576/Y (sky130_fd_sc_hd__o21ai_1)

0.08 0.00 1.84 v core/U581/A1 (sky130_fd_sc_hd__a21oi_1)

2 0.01 0.19 0.19 2.04 ^ core/U581/Y (sky130_fd_sc_hd__a21oi_1)

0.19 0.00 2.04 ^ core/U583/A2 (sky130_fd_sc_hd__o21ai_1)

2 0.01 0.08 0.11 2.15 v core/U583/Y (sky130_fd_sc_hd__o21ai_1)

0.08 0.00 2.15 v core/U587/A1 (sky130_fd_sc_hd__a21oi_1)

2 0.01 0.19 0.19 2.34 ^ core/U587/Y (sky130_fd_sc_hd__a21oi_1)

0.19 0.00 2.34 ^ core/U589/A2 (sky130_fd_sc_hd__o21ai_1)

2 0.01 0.08 0.11 2.45 v core/U589/Y (sky130_fd_sc_hd__o21ai_1)

0.08 0.00 2.45 v core/U344/A1 (sky130_fd_sc_hd__a21oi_1)

2 0.01 0.19 0.19 2.64 ^ core/U344/Y (sky130_fd_sc_hd__a21oi_1)

0.19 0.00 2.64 ^ core/U213/A2 (sky130_fd_sc_hd__o21ai_1)

2 0.01 0.08 0.11 2.75 v core/U213/Y (sky130_fd_sc_hd__o21ai_1)

0.08 0.00 2.75 v core/U343/A1 (sky130_fd_sc_hd__a21oi_1)

2 0.01 0.19 0.19 2.94 ^ core/U343/Y (sky130_fd_sc_hd__a21oi_1)

0.19 0.00 2.94 ^ core/U210/A2 (sky130_fd_sc_hd__o21ai_1)

2 0.01 0.08 0.11 3.05 v core/U210/Y (sky130_fd_sc_hd__o21ai_1)

0.08 0.00 3.05 v core/U342/A1 (sky130_fd_sc_hd__a21oi_1)

2 0.01 0.19 0.19 3.24 ^ core/U342/Y (sky130_fd_sc_hd__a21oi_1)

0.19 0.00 3.24 ^ core/U209/A2 (sky130_fd_sc_hd__o21ai_1)

2 0.01 0.08 0.11 3.35 v core/U209/Y (sky130_fd_sc_hd__o21ai_1)

0.08 0.00 3.35 v core/U341/A1 (sky130_fd_sc_hd__a21oi_1)

2 0.01 0.19 0.19 3.54 ^ core/U341/Y (sky130_fd_sc_hd__a21oi_1)

0.19 0.00 3.54 ^ core/U215/A2 (sky130_fd_sc_hd__o21ai_1)

2 0.01 0.08 0.11 3.65 v core/U215/Y (sky130_fd_sc_hd__o21ai_1)

0.08 0.00 3.65 v core/U96/A1 (sky130_fd_sc_hd__a21o_1)

1 0.00 0.04 0.17 3.82 v core/U96/X (sky130_fd_sc_hd__a21o_1)
 0.04 0.00 3.82 v core/U942/CIN (sky130_fd_sc_hd__fa_1)
 2 0.01 0.09 0.39 4.21 v core/U942/COUT (sky130_fd_sc_hd__fa_1)
 0.09 0.00 4.21 v core/U94/A1 (sky130_fd_sc_hd__a21o_1)
 1 0.00 0.04 0.18 4.39 v core/U94/X (sky130_fd_sc_hd__a21o_1)
 0.04 0.00 4.39 v core/U905/CIN (sky130_fd_sc_hd__fa_1)
 1 0.00 0.08 0.37 4.75 v core/U905/COUT (sky130_fd_sc_hd__fa_1)
 0.08 0.00 4.75 v core/U887/CIN (sky130_fd_sc_hd__fa_1)
 1 0.00 0.08 0.38 5.13 v core/U887/COUT (sky130_fd_sc_hd__fa_1)
 0.08 0.00 5.13 v core/U1370/CIN (sky130_fd_sc_hd__fa_1)
 1 0.01 0.08 0.38 5.52 v core/U1370/COUT (sky130_fd_sc_hd__fa_1)
 0.08 0.00 5.52 v core/U36/CIN (sky130_fd_sc_hd__fa_2)
 2 0.01 0.07 0.36 5.87 v core/U36/COUT (sky130_fd_sc_hd__fa_2)
 0.07 0.00 5.87 v core/U87/A (sky130_fd_sc_hd__clkinv_1)
 1 0.00 0.03 0.05 5.93 ^ core/U87/Y (sky130_fd_sc_hd__clkinv_1)
 0.03 0.00 5.93 ^ core/U613/A2 (sky130_fd_sc_hd__o21ai_1)
 1 0.00 0.06 0.06 5.98 v core/U613/Y (sky130_fd_sc_hd__o21ai_1)
 0.06 0.00 5.98 v core/U352/CIN (sky130_fd_sc_hd__fa_1)
 1 0.00 0.08 0.37 6.35 v core/U352/COUT (sky130_fd_sc_hd__fa_1)
 0.08 0.00 6.35 v core/U348/CIN (sky130_fd_sc_hd__fa_1)
 1 0.00 0.08 0.38 6.73 v core/U348/COUT (sky130_fd_sc_hd__fa_1)
 0.08 0.00 6.73 v core/U35/CIN (sky130_fd_sc_hd__fa_1)
 1 0.00 0.08 0.38 7.11 v core/U35/COUT (sky130_fd_sc_hd__fa_1)
 0.08 0.00 7.11 v core/U33/CIN (sky130_fd_sc_hd__fa_1)
 2 0.01 0.09 0.40 7.51 v core/U33/COUT (sky130_fd_sc_hd__fa_1)
 0.09 0.00 7.51 v core/U81/A1 (sky130_fd_sc_hd__a21o_1)
 1 0.00 0.04 0.18 7.69 v core/U81/X (sky130_fd_sc_hd__a21o_1)

```

0.04  0.00  7.69 v core/U347/CIN (sky130_fd_sc_hd__fa_1)
1  0.00  0.08  0.37  8.06 v core/U347/COUT (sky130_fd_sc_hd__fa_1)
0.08  0.00  8.06 v core/U351/CIN (sky130_fd_sc_hd__fa_1)
2  0.01  0.09  0.45  8.51 v core/U351/SUM (sky130_fd_sc_hd__fa_1)
0.09  0.00  8.51 v core/U3/A (sky130_fd_sc_hd__nand2_1)
15  0.04  0.35  0.29  8.80 ^ core/U3/Y (sky130_fd_sc_hd__nand2_1)
0.35  0.00  8.80 ^ core/U684/B2 (sky130_fd_sc_hd__o22ai_1)
1  0.00  0.14  0.10  8.90 v core/U684/Y (sky130_fd_sc_hd__o22ai_1)
0.14  0.00  8.90 v core/CPU_Xreg_value_a4_reg[27][30]/D (sky130_fd_sc_hd__dfxtp_1)
8.90  data arrival time

```

```

0.00  11.00  11.00  clock clk (rise edge)
0.00  11.00  clock network delay (ideal)
0.00  11.00  clock reconvergence pessimism
11.00 ^ core/CPU_Xreg_value_a4_reg[27][30]/CLK (sky130_fd_sc_hd__dfxtp_1)
-0.16  10.84  library setup time
10.84  data required time

```

```

10.84  data required time
-8.90  data arrival time

```

```

1.94  slack (MET)

```

----- WNS -----

wns max 0.00

----- TNS -----

tns max 0.00

=====

END OF LIBRARY: sky130_fd_sc_hd__tt_025C_1v80.lib

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LIBRARY: sky130_fd_sc_hd__tt_100C_1v80.lib

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Warning 1140:
/data/Desktop/work/tools/openlane_working_dir/OpenSTA/vsdbabysoc/lib/sky130_fd_sc_hd__tt_100C_1v80.lib line 1, library sky130_fd_sc_hd__tt_100C_1v80 already exists.

Warning 1140:
/data/Desktop/work/tools/openlane_working_dir/OpenSTA/vsdbabysoc/lib/avsddac.lib line 1, library avsddac already exists.

Warning 1140:
/data/Desktop/work/tools/openlane_working_dir/OpenSTA/vsdbabysoc/lib/avsddac.lib line 1, library avsddac already exists.

Warning 1140:
/data/Desktop/work/tools/openlane_working_dir/OpenSTA/vsdbabysoc/lib/avsdpll.lib line 1, library avsdpll already exists.

Warning 1140:
/data/Desktop/work/tools/openlane_working_dir/OpenSTA/vsdbabysoc/lib/avsdpll.lib line 1, library avsdpll already exists.

----- REPORT CHECKS -----

Warning 168: run_final.tcl line 1, unknown field nets.

Startpoint: core/CPU_imm_a2_reg[10]

(rising edge-triggered flip-flop clocked by clk)

Endpoint: core/CPU_imm_a3_reg[10]

(rising edge-triggered flip-flop clocked by clk)

Path Group: clk

Path Type: min

Fanout	Cap	Slew	Delay	Time	Description
--------	-----	------	-------	------	-------------

0.00	0.00	0.00	0.00		clock clk (rise edge)
------	------	------	------	--	-----------------------

0.00	0.00	0.00	0.00		clock network delay (ideal)
------	------	------	------	--	-----------------------------

0.00	0.00	0.00	0.00		^ core/CPU_imm_a2_reg[10]/CLK (sky130_fd_sc_hd__dfxtp_1)
------	------	------	------	--	--

1	0.00	0.03	0.28	0.28	^ core/CPU_imm_a2_reg[10]/Q (sky130_fd_sc_hd__dfxtp_1)
---	------	------	------	------	--

0.03	0.00	0.28	0.28		^ core/CPU_imm_a3_reg[10]/D (sky130_fd_sc_hd__dfxtp_1)
------	------	------	------	--	--

0.28					data arrival time
------	--	--	--	--	-------------------

0.00	0.00	0.00	0.00		clock clk (rise edge)
------	------	------	------	--	-----------------------

0.00	0.00	0.00	0.00		clock network delay (ideal)
------	------	------	------	--	-----------------------------

0.00	0.00	0.00	0.00		clock reconvergence pessimism
------	------	------	------	--	-------------------------------

0.00					^ core/CPU_imm_a3_reg[10]/CLK (sky130_fd_sc_hd__dfxtp_1)
------	--	--	--	--	--

-0.04	-0.04				library hold time
-------	-------	--	--	--	-------------------

-0.04					data required time
-------	--	--	--	--	--------------------

-0.04					data required time
-------	--	--	--	--	--------------------

-0.28					data arrival time
-------	--	--	--	--	-------------------

0.31					slack (MET)
------	--	--	--	--	-------------

Startpoint: core/CPU_is_addi_a3_reg

(rising edge-triggered flip-flop clocked by clk)

Endpoint: core/CPU_Xreg_value_a4_reg[27][30]

(rising edge-triggered flip-flop clocked by clk)

Path Group: clk

Path Type: max

Fanout	Cap	Slew	Delay	Time	Description
--------	-----	------	-------	------	-------------

	0.00	0.00	0.00		clock clk (rise edge)
	0.00	0.00			clock network delay (ideal)
	0.00	0.00	0.00		^ core/CPU_is_addi_a3_reg/CLK (sky130_fd_sc_hd__dfxtp_1)
2	0.01	0.04	0.29	0.29	v core/CPU_is_addi_a3_reg/Q (sky130_fd_sc_hd__dfxtp_1)
	0.04	0.00	0.29		v core/U474/A (sky130_fd_sc_hd__nor2_1)
2	0.00	0.11	0.11	0.40	^ core/U474/Y (sky130_fd_sc_hd__nor2_1)
	0.11	0.00	0.40		^ core/U476/A (sky130_fd_sc_hd__nand2_1)
2	0.01	0.09	0.10	0.51	v core/U476/Y (sky130_fd_sc_hd__nand2_1)
	0.09	0.00	0.51		v core/U9/A (sky130_fd_sc_hd__inv_2)
34	0.15	0.64	0.48	0.99	^ core/U9/Y (sky130_fd_sc_hd__inv_2)
	0.64	0.00	0.99		^ core/U479/B (sky130_fd_sc_hd__xor2_1)
2	0.01	0.18	0.19	1.18	v core/U479/X (sky130_fd_sc_hd__xor2_1)
	0.18	0.00	1.18		v core/U569/A2 (sky130_fd_sc_hd__a21oi_1)
2	0.01	0.19	0.23	1.41	^ core/U569/Y (sky130_fd_sc_hd__a21oi_1)
	0.19	0.00	1.41		^ core/U571/A2 (sky130_fd_sc_hd__o21ai_1)
2	0.01	0.09	0.12	1.53	v core/U571/Y (sky130_fd_sc_hd__o21ai_1)

0.09 0.00 1.53 v core/U574/A1 (sky130_fd_sc_hd__a21oi_1)

2 0.01 0.19 0.18 1.71 ^ core/U574/Y (sky130_fd_sc_hd__a21oi_1)

0.19 0.00 1.71 ^ core/U576/A2 (sky130_fd_sc_hd__o21ai_1)

2 0.01 0.09 0.11 1.83 v core/U576/Y (sky130_fd_sc_hd__o21ai_1)

0.09 0.00 1.83 v core/U581/A1 (sky130_fd_sc_hd__a21oi_1)

2 0.01 0.19 0.18 2.01 ^ core/U581/Y (sky130_fd_sc_hd__a21oi_1)

0.19 0.00 2.01 ^ core/U583/A2 (sky130_fd_sc_hd__o21ai_1)

2 0.01 0.09 0.11 2.12 v core/U583/Y (sky130_fd_sc_hd__o21ai_1)

0.09 0.00 2.12 v core/U587/A1 (sky130_fd_sc_hd__a21oi_1)

2 0.01 0.19 0.18 2.31 ^ core/U587/Y (sky130_fd_sc_hd__a21oi_1)

0.19 0.00 2.31 ^ core/U589/A2 (sky130_fd_sc_hd__o21ai_1)

2 0.01 0.09 0.11 2.42 v core/U589/Y (sky130_fd_sc_hd__o21ai_1)

0.09 0.00 2.42 v core/U344/A1 (sky130_fd_sc_hd__a21oi_1)

2 0.01 0.19 0.18 2.61 ^ core/U344/Y (sky130_fd_sc_hd__a21oi_1)

0.19 0.00 2.61 ^ core/U213/A2 (sky130_fd_sc_hd__o21ai_1)

2 0.01 0.09 0.11 2.72 v core/U213/Y (sky130_fd_sc_hd__o21ai_1)

0.09 0.00 2.72 v core/U343/A1 (sky130_fd_sc_hd__a21oi_1)

2 0.01 0.19 0.18 2.90 ^ core/U343/Y (sky130_fd_sc_hd__a21oi_1)

0.19 0.00 2.90 ^ core/U210/A2 (sky130_fd_sc_hd__o21ai_1)

2 0.01 0.09 0.11 3.02 v core/U210/Y (sky130_fd_sc_hd__o21ai_1)

0.09 0.00 3.02 v core/U342/A1 (sky130_fd_sc_hd__a21oi_1)

2 0.01 0.19 0.18 3.20 ^ core/U342/Y (sky130_fd_sc_hd__a21oi_1)

0.19 0.00 3.20 ^ core/U209/A2 (sky130_fd_sc_hd__o21ai_1)

2 0.01 0.09 0.11 3.32 v core/U209/Y (sky130_fd_sc_hd__o21ai_1)

0.09 0.00 3.32 v core/U341/A1 (sky130_fd_sc_hd__a21oi_1)

2 0.01 0.19 0.18 3.50 ^ core/U341/Y (sky130_fd_sc_hd__a21oi_1)

0.19 0.00 3.50 ^ core/U215/A2 (sky130_fd_sc_hd__o21ai_1)

2 0.01 0.09 0.11 3.62 v core/U215/Y (sky130_fd_sc_hd__o21ai_1)
 0.09 0.00 3.62 v core/U96/A1 (sky130_fd_sc_hd__a21o_1)
 1 0.00 0.05 0.17 3.79 v core/U96/X (sky130_fd_sc_hd__a21o_1)
 0.05 0.00 3.79 v core/U942/CIN (sky130_fd_sc_hd__fa_1)
 2 0.01 0.10 0.38 4.16 v core/U942/COUT (sky130_fd_sc_hd__fa_1)
 0.10 0.00 4.16 v core/U94/A1 (sky130_fd_sc_hd__a21o_1)
 1 0.00 0.05 0.17 4.33 v core/U94/X (sky130_fd_sc_hd__a21o_1)
 0.05 0.00 4.33 v core/U905/CIN (sky130_fd_sc_hd__fa_1)
 1 0.00 0.08 0.36 4.69 v core/U905/COUT (sky130_fd_sc_hd__fa_1)
 0.08 0.00 4.69 v core/U887/CIN (sky130_fd_sc_hd__fa_1)
 1 0.00 0.08 0.37 5.06 v core/U887/COUT (sky130_fd_sc_hd__fa_1)
 0.08 0.00 5.06 v core/U1370/CIN (sky130_fd_sc_hd__fa_1)
 1 0.01 0.09 0.37 5.43 v core/U1370/COUT (sky130_fd_sc_hd__fa_1)
 0.09 0.00 5.43 v core/U36/CIN (sky130_fd_sc_hd__fa_2)
 2 0.01 0.08 0.35 5.78 v core/U36/COUT (sky130_fd_sc_hd__fa_2)
 0.08 0.00 5.78 v core/U87/A (sky130_fd_sc_hd__clkinv_1)
 1 0.00 0.03 0.05 5.83 ^ core/U87/Y (sky130_fd_sc_hd__clkinv_1)
 0.03 0.00 5.83 ^ core/U613/A2 (sky130_fd_sc_hd__o21ai_1)
 1 0.00 0.07 0.06 5.89 v core/U613/Y (sky130_fd_sc_hd__o21ai_1)
 0.07 0.00 5.89 v core/U352/CIN (sky130_fd_sc_hd__fa_1)
 1 0.00 0.08 0.36 6.26 v core/U352/COUT (sky130_fd_sc_hd__fa_1)
 0.08 0.00 6.26 v core/U348/CIN (sky130_fd_sc_hd__fa_1)
 1 0.00 0.08 0.37 6.62 v core/U348/COUT (sky130_fd_sc_hd__fa_1)
 0.08 0.00 6.62 v core/U35/CIN (sky130_fd_sc_hd__fa_1)
 1 0.00 0.08 0.37 6.99 v core/U35/COUT (sky130_fd_sc_hd__fa_1)
 0.08 0.00 6.99 v core/U33/CIN (sky130_fd_sc_hd__fa_1)
 2 0.01 0.10 0.39 7.38 v core/U33/COUT (sky130_fd_sc_hd__fa_1)

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0.10  0.00  7.38 v core/U81/A1 (sky130_fd_sc_hd__a21o_1)
1  0.00  0.05  0.17  7.55 v core/U81/X (sky130_fd_sc_hd__a21o_1)
0.05  0.00  7.55 v core/U347/CIN (sky130_fd_sc_hd__fa_1)
1  0.00  0.08  0.36  7.91 v core/U347/COUT (sky130_fd_sc_hd__fa_1)
0.08  0.00  7.91 v core/U351/CIN (sky130_fd_sc_hd__fa_1)
2  0.01  0.08  0.49  8.40 ^ core/U351/SUM (sky130_fd_sc_hd__fa_1)
0.08  0.00  8.40 ^ core/U3/A (sky130_fd_sc_hd__nand2_1)
15  0.03  0.34  0.25  8.65 v core/U3/Y (sky130_fd_sc_hd__nand2_1)
0.34  0.00  8.65 v core/U684/B2 (sky130_fd_sc_hd__o22ai_1)
1  0.00  0.20  0.16  8.81 ^ core/U684/Y (sky130_fd_sc_hd__o22ai_1)
0.20  0.00  8.81 ^ core/CPU_Xreg_value_a4_reg[27][30]/D (sky130_fd_sc_hd__dfxtp_1)
8.81  data arrival time

```

```

0.00  11.00  11.00 clock clk (rise edge)
0.00  11.00 clock network delay (ideal)
0.00  11.00 clock reconvergence pessimism
11.00 ^ core/CPU_Xreg_value_a4_reg[27][30]/CLK (sky130_fd_sc_hd__dfxtp_1)
-0.10  10.90 library setup time
10.90  data required time

```

```

10.90  data required time
-8.81  data arrival time

```

```

2.09  slack (MET)

```

----- WNS -----

wns max 0.00

----- TNS -----

tns max 0.00

=====

END OF LIBRARY: sky130_fd_sc_hd__tt_100C_1v80.lib

=====

=====

LIBRARY: sky130_fd_sc_hd__ss_n40C_1v28.lib

=====

Warning 1140:
/data/Desktop/work/tools/openlane_working_dir/OpenSTA/vsdbabysoc/lib/sky130_fd_sc_hd__ss_n40C_1v28.lib line 1, library sky130_fd_sc_hd__ss_n40C_1v28 already exists.

Warning 1140:
/data/Desktop/work/tools/openlane_working_dir/OpenSTA/vsdbabysoc/lib/avsdac.lib line 1, library avsdac already exists.

Warning 1140:
/data/Desktop/work/tools/openlane_working_dir/OpenSTA/vsdbabysoc/lib/avsdac.lib line 1, library avsdac already exists.

Warning 1140:
/data/Desktop/work/tools/openlane_working_dir/OpenSTA/vsdbabysoc/lib/avsdpll.lib line 1, library avsdpll already exists.

Warning 1140:
/data/Desktop/work/tools/openlane_working_dir/OpenSTA/vsdbabysoc/lib/avsdpll.lib line 1, library avsdpll already exists.

Fanout	Cap	Slew	Delay	Time	Description

	0.00	0.00	0.00		clock clk (rise edge)
		0.00	0.00		clock network delay (ideal)
	0.00	0.00	0.00		^ core/CPU_imm_a2_reg[10]/CLK (sky130_fd_sc_hd__dfxtp_1)
1	0.00	0.07	1.45	1.45	v core/CPU_imm_a2_reg[10]/Q (sky130_fd_sc_hd__dfxtp_1)
	0.07	0.00	1.45		v core/CPU_imm_a3_reg[10]/D (sky130_fd_sc_hd__dfxtp_1)
			1.45		data arrival time
	0.00	0.00	0.00		clock clk (rise edge)
		0.00	0.00		clock network delay (ideal)
	0.00	0.00			clock reconvergence pessimism
		0.00			^ core/CPU_imm_a3_reg[10]/CLK (sky130_fd_sc_hd__dfxtp_1)
	-0.38	-0.38			library hold time
		-0.38			data required time

		-0.38			data required time
		-1.45			data arrival time

1.83 slack (MET)

Startpoint: core/CPU_is_addi_a3_reg

(rising edge-triggered flip-flop clocked by clk)

Endpoint: core/CPU_Xreg_value_a4_reg[27][30]

(rising edge-triggered flip-flop clocked by clk)

Path Group: clk

Path Type: max

Fanout	Cap	Slew	Delay	Time	Description
--------	-----	------	-------	------	-------------

	0.00	0.00	0.00		clock clk (rise edge)
	0.00	0.00			clock network delay (ideal)
	0.00	0.00	0.00		^ core/CPU_is_addi_a3_reg/CLK (sky130_fd_sc_hd__dfxtp_1)
2	0.00	0.11	1.50	1.50	v core/CPU_is_addi_a3_reg/Q (sky130_fd_sc_hd__dfxtp_1)
	0.11	0.00	1.50		v core/U474/A (sky130_fd_sc_hd__nor2_1)
2	0.00	0.65	0.71	2.20	^ core/U474/Y (sky130_fd_sc_hd__nor2_1)
	0.65	0.00	2.20		^ core/U476/A (sky130_fd_sc_hd__nand2_1)
2	0.01	0.23	0.56	2.76	v core/U476/Y (sky130_fd_sc_hd__nand2_1)
	0.23	0.00	2.76		v core/U9/A (sky130_fd_sc_hd__inv_2)
34	0.13	3.38	2.67	5.43	^ core/U9/Y (sky130_fd_sc_hd__inv_2)
	3.38	0.00	5.43		^ core/U479/B (sky130_fd_sc_hd__xor2_1)
2	0.01	0.53	1.71	7.14	v core/U479/X (sky130_fd_sc_hd__xor2_1)
	0.53	0.00	7.14		v core/U569/A2 (sky130_fd_sc_hd__a21oi_1)
2	0.01	1.07	1.42	8.57	^ core/U569/Y (sky130_fd_sc_hd__a21oi_1)

1.07 0.00 8.57 ^ core/U571/A2 (sky130_fd_sc_hd__o21ai_1)

2 0.01 0.22 0.76 9.33 v core/U571/Y (sky130_fd_sc_hd__o21ai_1)

0.22 0.00 9.33 v core/U574/A1 (sky130_fd_sc_hd__a21oi_1)

2 0.01 1.07 1.20 10.53 ^ core/U574/Y (sky130_fd_sc_hd__a21oi_1)

1.07 0.00 10.53 ^ core/U576/A2 (sky130_fd_sc_hd__o21ai_1)

2 0.01 0.22 0.76 11.29 v core/U576/Y (sky130_fd_sc_hd__o21ai_1)

0.22 0.00 11.29 v core/U581/A1 (sky130_fd_sc_hd__a21oi_1)

2 0.01 1.07 1.20 12.49 ^ core/U581/Y (sky130_fd_sc_hd__a21oi_1)

1.07 0.00 12.49 ^ core/U583/A2 (sky130_fd_sc_hd__o21ai_1)

2 0.01 0.22 0.76 13.25 v core/U583/Y (sky130_fd_sc_hd__o21ai_1)

0.22 0.00 13.25 v core/U587/A1 (sky130_fd_sc_hd__a21oi_1)

2 0.01 1.07 1.20 14.45 ^ core/U587/Y (sky130_fd_sc_hd__a21oi_1)

1.07 0.00 14.45 ^ core/U589/A2 (sky130_fd_sc_hd__o21ai_1)

2 0.01 0.22 0.76 15.21 v core/U589/Y (sky130_fd_sc_hd__o21ai_1)

0.22 0.00 15.21 v core/U344/A1 (sky130_fd_sc_hd__a21oi_1)

2 0.01 1.07 1.20 16.41 ^ core/U344/Y (sky130_fd_sc_hd__a21oi_1)

1.07 0.00 16.41 ^ core/U213/A2 (sky130_fd_sc_hd__o21ai_1)

2 0.01 0.22 0.76 17.17 v core/U213/Y (sky130_fd_sc_hd__o21ai_1)

0.22 0.00 17.17 v core/U343/A1 (sky130_fd_sc_hd__a21oi_1)

2 0.01 1.07 1.20 18.38 ^ core/U343/Y (sky130_fd_sc_hd__a21oi_1)

1.07 0.00 18.38 ^ core/U210/A2 (sky130_fd_sc_hd__o21ai_1)

2 0.01 0.22 0.76 19.14 v core/U210/Y (sky130_fd_sc_hd__o21ai_1)

0.22 0.00 19.14 v core/U342/A1 (sky130_fd_sc_hd__a21oi_1)

2 0.01 1.07 1.20 20.34 ^ core/U342/Y (sky130_fd_sc_hd__a21oi_1)

1.07 0.00 20.34 ^ core/U209/A2 (sky130_fd_sc_hd__o21ai_1)

2 0.01 0.22 0.76 21.10 v core/U209/Y (sky130_fd_sc_hd__o21ai_1)

0.22 0.00 21.10 v core/U341/A1 (sky130_fd_sc_hd__a21oi_1)

2 0.01 1.07 1.20 22.30 ^ core/U341/Y (sky130_fd_sc_hd__a21oi_1)
 1.07 0.00 22.30 ^ core/U215/A2 (sky130_fd_sc_hd__o21ai_1)
 2 0.01 0.22 0.76 23.06 v core/U215/Y (sky130_fd_sc_hd__o21ai_1)
 0.22 0.00 23.06 v core/U96/A1 (sky130_fd_sc_hd__a21o_1)
 1 0.00 0.15 1.13 24.19 v core/U96/X (sky130_fd_sc_hd__a21o_1)
 0.15 0.00 24.19 v core/U942/CIN (sky130_fd_sc_hd__fa_1)
 2 0.01 0.49 3.70 27.89 v core/U942/COUT (sky130_fd_sc_hd__fa_1)
 0.49 0.00 27.89 v core/U94/A1 (sky130_fd_sc_hd__a21o_1)
 1 0.00 0.15 1.32 29.21 v core/U94/X (sky130_fd_sc_hd__a21o_1)
 0.15 0.00 29.21 v core/U905/CIN (sky130_fd_sc_hd__fa_1)
 1 0.00 0.45 3.62 32.83 v core/U905/COUT (sky130_fd_sc_hd__fa_1)
 0.45 0.00 32.83 v core/U887/CIN (sky130_fd_sc_hd__fa_1)
 1 0.00 0.45 3.78 36.61 v core/U887/COUT (sky130_fd_sc_hd__fa_1)
 0.45 0.00 36.61 v core/U1370/CIN (sky130_fd_sc_hd__fa_1)
 1 0.00 0.46 3.80 40.42 v core/U1370/COUT (sky130_fd_sc_hd__fa_1)
 0.46 0.00 40.42 v core/U36/CIN (sky130_fd_sc_hd__fa_2)
 2 0.01 0.44 3.39 43.81 v core/U36/COUT (sky130_fd_sc_hd__fa_2)
 0.44 0.00 43.81 v core/U87/A (sky130_fd_sc_hd__clkinv_1)
 1 0.00 0.12 0.40 44.21 ^ core/U87/Y (sky130_fd_sc_hd__clkinv_1)
 0.12 0.00 44.21 ^ core/U613/A2 (sky130_fd_sc_hd__o21ai_1)
 1 0.00 0.17 0.21 44.43 v core/U613/Y (sky130_fd_sc_hd__o21ai_1)
 0.17 0.00 44.43 v core/U352/CIN (sky130_fd_sc_hd__fa_1)
 1 0.00 0.45 3.63 48.06 v core/U352/COUT (sky130_fd_sc_hd__fa_1)
 0.45 0.00 48.06 v core/U348/CIN (sky130_fd_sc_hd__fa_1)
 1 0.00 0.45 3.78 51.84 v core/U348/COUT (sky130_fd_sc_hd__fa_1)
 0.45 0.00 51.84 v core/U35/CIN (sky130_fd_sc_hd__fa_1)
 1 0.00 0.45 3.78 55.62 v core/U35/COUT (sky130_fd_sc_hd__fa_1)

```

0.45  0.00  55.62 v core/U33/CIN (sky130_fd_sc_hd__fa_1)
2  0.01  0.49  3.86  59.48 v core/U33/COUT (sky130_fd_sc_hd__fa_1)
0.49  0.00  59.48 v core/U81/A1 (sky130_fd_sc_hd__a21o_1)
1  0.00  0.15  1.32  60.80 v core/U81/X (sky130_fd_sc_hd__a21o_1)
0.15  0.00  60.80 v core/U347/CIN (sky130_fd_sc_hd__fa_1)
1  0.00  0.45  3.62  64.42 v core/U347/COUT (sky130_fd_sc_hd__fa_1)
0.45  0.00  64.42 v core/U351/CIN (sky130_fd_sc_hd__fa_1)
2  0.00  0.51  4.76  69.19 v core/U351/SUM (sky130_fd_sc_hd__fa_1)
0.51  0.00  69.19 v core/U3/A (sky130_fd_sc_hd__nand2_1)
15  0.03  1.70  1.61  70.80 ^ core/U3/Y (sky130_fd_sc_hd__nand2_1)
1.70  0.00  70.80 ^ core/U684/B2 (sky130_fd_sc_hd__o22ai_1)
1  0.00  0.42  0.91  71.70 v core/U684/Y (sky130_fd_sc_hd__o22ai_1)
0.42  0.00  71.70 v core/CPU_Xreg_value_a4_reg[27][30]/D (sky130_fd_sc_hd__dfxtp_1)
71.70  data arrival time

```

```

0.00  11.00  11.00  clock clk (rise edge)
0.00  11.00  clock network delay (ideal)
0.00  11.00  clock reconvergence pessimism
11.00 ^ core/CPU_Xreg_value_a4_reg[27][30]/CLK (sky130_fd_sc_hd__dfxtp_1)
-1.86  9.14  library setup time
9.14  data required time

```

```

-----
9.14  data required time

```

```

-71.70  data arrival time

```

```

-----
-62.56  slack (VIOLATED)

```

----- WNS -----

wns max -62.56

----- TNS -----

tns max -13957.59

=====

END OF LIBRARY: sky130_fd_sc_hd__ss_n40C_1v28.lib

=====

=====

LIBRARY: sky130_fd_sc_hd__ss_n40C_1v35.lib

=====

Warning 1140:
/data/Desktop/work/tools/openlane_working_dir/OpenSTA/vsdbabysoc/lib/sky130_fd_sc_hd__ss_n40C_1v35.lib line 1, library sky130_fd_sc_hd__ss_n40C_1v35 already exists.

Warning 1140:
/data/Desktop/work/tools/openlane_working_dir/OpenSTA/vsdbabysoc/lib/avsddac.lib line 1, library avsddac already exists.

Warning 1140:
/data/Desktop/work/tools/openlane_working_dir/OpenSTA/vsdbabysoc/lib/avsddac.lib line 1, library avsddac already exists.

Warning 1140:
/data/Desktop/work/tools/openlane_working_dir/OpenSTA/vsdbabysoc/lib/avsdpll.lib line 1, library avsdpll already exists.

Warning 1140:
/data/Desktop/work/tools/openlane_working_dir/OpenSTA/vsdbabysoc/lib/avsdpll.lib line 1, library
avsdpll already exists.

----- REPORT CHECKS -----

Warning 168: run_final.tcl line 1, unknown field nets.

Startpoint: core/CPU_imm_a2_reg[10]

(rising edge-triggered flip-flop clocked by clk)

Endpoint: core/CPU_imm_a3_reg[10]

(rising edge-triggered flip-flop clocked by clk)

Path Group: clk

Path Type: min

Fanout	Cap	Slew	Delay	Time	Description
--------	-----	------	-------	------	-------------

	0.00	0.00	0.00		clock clk (rise edge)
		0.00	0.00		clock network delay (ideal)
	0.00	0.00	0.00		^ core/CPU_imm_a2_reg[10]/CLK (sky130_fd_sc_hd__dfxtp_1)
1	0.00	0.06	1.08	1.08	v core/CPU_imm_a2_reg[10]/Q (sky130_fd_sc_hd__dfxtp_1)
	0.06	0.00	1.08		v core/CPU_imm_a3_reg[10]/D (sky130_fd_sc_hd__dfxtp_1)
			1.08		data arrival time
	0.00	0.00	0.00		clock clk (rise edge)
		0.00	0.00		clock network delay (ideal)
		0.00	0.00		clock reconvergence pessimism
			0.00		^ core/CPU_imm_a3_reg[10]/CLK (sky130_fd_sc_hd__dfxtp_1)
	-0.27	-0.27			library hold time
		-0.27			data required time

-0.27 data required time

-1.08 data arrival time

1.35 slack (MET)

Startpoint: core/CPU_is_addi_a3_reg

(rising edge-triggered flip-flop clocked by clk)

Endpoint: core/CPU_Xreg_value_a4_reg[27][30]

(rising edge-triggered flip-flop clocked by clk)

Path Group: clk

Path Type: max

Fanout	Cap	Slew	Delay	Time	Description
--------	-----	------	-------	------	-------------

0.00	0.00	0.00	0.00		clock clk (rise edge)
------	------	------	------	--	-----------------------

0.00	0.00	0.00	0.00		clock network delay (ideal)
------	------	------	------	--	-----------------------------

0.00	0.00	0.00	0.00		^ core/CPU_is_addi_a3_reg/CLK (sky130_fd_sc_hd__dfxtp_1)
------	------	------	------	--	--

2	0.00	0.09	1.12	1.12	v core/CPU_is_addi_a3_reg/Q (sky130_fd_sc_hd__dfxtp_1)
---	------	------	------	------	--

0.09	0.00	1.12	1.12		v core/U474/A (sky130_fd_sc_hd__nor2_1)
------	------	------	------	--	---

2	0.00	0.47	0.52	1.64	^ core/U474/Y (sky130_fd_sc_hd__nor2_1)
---	------	------	------	------	---

0.47	0.00	1.64	1.64		^ core/U476/A (sky130_fd_sc_hd__nand2_1)
------	------	------	------	--	--

2	0.01	0.19	0.41	2.05	v core/U476/Y (sky130_fd_sc_hd__nand2_1)
---	------	------	------	------	--

0.19	0.00	2.05	2.05		v core/U9/A (sky130_fd_sc_hd__inv_2)
------	------	------	------	--	--------------------------------------

34	0.13	2.53	2.01	4.06	^ core/U9/Y (sky130_fd_sc_hd__inv_2)
----	------	------	------	------	--------------------------------------

2.53	0.00	4.06	4.06		^ core/U479/B (sky130_fd_sc_hd__xor2_1)
------	------	------	------	--	---

2 0.01 0.41 1.22 5.28 v core/U479/X (sky130_fd_sc_hd__xor2_1)
 0.41 0.00 5.28 v core/U569/A2 (sky130_fd_sc_hd__a21oi_1)
 2 0.01 0.77 1.04 6.33 ^ core/U569/Y (sky130_fd_sc_hd__a21oi_1)
 0.77 0.00 6.33 ^ core/U571/A2 (sky130_fd_sc_hd__o21ai_1)
 2 0.01 0.17 0.54 6.87 v core/U571/Y (sky130_fd_sc_hd__o21ai_1)
 0.17 0.00 6.87 v core/U574/A1 (sky130_fd_sc_hd__a21oi_1)
 2 0.01 0.77 0.87 7.74 ^ core/U574/Y (sky130_fd_sc_hd__a21oi_1)
 0.77 0.00 7.74 ^ core/U576/A2 (sky130_fd_sc_hd__o21ai_1)
 2 0.01 0.17 0.54 8.28 v core/U576/Y (sky130_fd_sc_hd__o21ai_1)
 0.17 0.00 8.28 v core/U581/A1 (sky130_fd_sc_hd__a21oi_1)
 2 0.01 0.77 0.87 9.15 ^ core/U581/Y (sky130_fd_sc_hd__a21oi_1)
 0.77 0.00 9.15 ^ core/U583/A2 (sky130_fd_sc_hd__o21ai_1)
 2 0.01 0.17 0.54 9.69 v core/U583/Y (sky130_fd_sc_hd__o21ai_1)
 0.17 0.00 9.69 v core/U587/A1 (sky130_fd_sc_hd__a21oi_1)
 2 0.01 0.77 0.87 10.56 ^ core/U587/Y (sky130_fd_sc_hd__a21oi_1)
 0.77 0.00 10.56 ^ core/U589/A2 (sky130_fd_sc_hd__o21ai_1)
 2 0.01 0.17 0.54 11.10 v core/U589/Y (sky130_fd_sc_hd__o21ai_1)
 0.17 0.00 11.10 v core/U344/A1 (sky130_fd_sc_hd__a21oi_1)
 2 0.01 0.77 0.87 11.96 ^ core/U344/Y (sky130_fd_sc_hd__a21oi_1)
 0.77 0.00 11.96 ^ core/U213/A2 (sky130_fd_sc_hd__o21ai_1)
 2 0.01 0.17 0.54 12.51 v core/U213/Y (sky130_fd_sc_hd__o21ai_1)
 0.17 0.00 12.51 v core/U343/A1 (sky130_fd_sc_hd__a21oi_1)
 2 0.01 0.77 0.87 13.37 ^ core/U343/Y (sky130_fd_sc_hd__a21oi_1)
 0.77 0.00 13.37 ^ core/U210/A2 (sky130_fd_sc_hd__o21ai_1)
 2 0.01 0.17 0.54 13.92 v core/U210/Y (sky130_fd_sc_hd__o21ai_1)
 0.17 0.00 13.92 v core/U342/A1 (sky130_fd_sc_hd__a21oi_1)
 2 0.01 0.77 0.87 14.78 ^ core/U342/Y (sky130_fd_sc_hd__a21oi_1)

0.77 0.00 14.78 ^ core/U209/A2 (sky130_fd_sc_hd__o21ai_1)
 2 0.01 0.17 0.54 15.33 v core/U209/Y (sky130_fd_sc_hd__o21ai_1)
 0.17 0.00 15.33 v core/U341/A1 (sky130_fd_sc_hd__a21oi_1)
 2 0.01 0.77 0.87 16.19 ^ core/U341/Y (sky130_fd_sc_hd__a21oi_1)
 0.77 0.00 16.19 ^ core/U215/A2 (sky130_fd_sc_hd__o21ai_1)
 2 0.01 0.17 0.54 16.74 v core/U215/Y (sky130_fd_sc_hd__o21ai_1)
 0.17 0.00 16.74 v core/U96/A1 (sky130_fd_sc_hd__a21o_1)
 1 0.00 0.11 0.81 17.54 v core/U96/X (sky130_fd_sc_hd__a21o_1)
 0.11 0.00 17.54 v core/U942/CIN (sky130_fd_sc_hd__fa_1)
 2 0.01 0.34 2.47 20.01 v core/U942/COUT (sky130_fd_sc_hd__fa_1)
 0.34 0.00 20.01 v core/U94/A1 (sky130_fd_sc_hd__a21o_1)
 1 0.00 0.11 0.91 20.92 v core/U94/X (sky130_fd_sc_hd__a21o_1)
 0.11 0.00 20.92 v core/U905/CIN (sky130_fd_sc_hd__fa_1)
 1 0.00 0.31 2.42 23.34 v core/U905/COUT (sky130_fd_sc_hd__fa_1)
 0.31 0.00 23.34 v core/U887/CIN (sky130_fd_sc_hd__fa_1)
 1 0.00 0.31 2.52 25.87 v core/U887/COUT (sky130_fd_sc_hd__fa_1)
 0.31 0.00 25.87 v core/U1370/CIN (sky130_fd_sc_hd__fa_1)
 1 0.00 0.32 2.54 28.40 v core/U1370/COUT (sky130_fd_sc_hd__fa_1)
 0.32 0.00 28.40 v core/U36/CIN (sky130_fd_sc_hd__fa_2)
 2 0.01 0.30 2.27 30.67 v core/U36/COUT (sky130_fd_sc_hd__fa_2)
 0.30 0.00 30.67 v core/U87/A (sky130_fd_sc_hd__clkinv_1)
 1 0.00 0.09 0.27 30.94 ^ core/U87/Y (sky130_fd_sc_hd__clkinv_1)
 0.09 0.00 30.94 ^ core/U613/A2 (sky130_fd_sc_hd__o21ai_1)
 1 0.00 0.14 0.17 31.11 v core/U613/Y (sky130_fd_sc_hd__o21ai_1)
 0.14 0.00 31.11 v core/U352/CIN (sky130_fd_sc_hd__fa_1)
 1 0.00 0.31 2.43 33.54 v core/U352/COUT (sky130_fd_sc_hd__fa_1)
 0.31 0.00 33.54 v core/U348/CIN (sky130_fd_sc_hd__fa_1)

```

1  0.00  0.31  2.52  36.07 v core/U348/COUT (sky130_fd_sc_hd__fa_1)
    0.31  0.00  36.07 v core/U35/CIN (sky130_fd_sc_hd__fa_1)
1  0.00  0.31  2.52  38.59 v core/U35/COUT (sky130_fd_sc_hd__fa_1)
    0.31  0.00  38.59 v core/U33/CIN (sky130_fd_sc_hd__fa_1)
2  0.01  0.34  2.57  41.16 v core/U33/COUT (sky130_fd_sc_hd__fa_1)
    0.34  0.00  41.16 v core/U81/A1 (sky130_fd_sc_hd__a21o_1)
1  0.00  0.11  0.91  42.07 v core/U81/X (sky130_fd_sc_hd__a21o_1)
    0.11  0.00  42.07 v core/U347/CIN (sky130_fd_sc_hd__fa_1)
1  0.00  0.31  2.42  44.49 v core/U347/COUT (sky130_fd_sc_hd__fa_1)
    0.31  0.00  44.49 v core/U351/CIN (sky130_fd_sc_hd__fa_1)
2  0.00  0.35  3.14  47.63 v core/U351/SUM (sky130_fd_sc_hd__fa_1)
    0.35  0.00  47.63 v core/U3/A (sky130_fd_sc_hd__nand2_1)
15 0.03  1.29  1.20  48.83 ^ core/U3/Y (sky130_fd_sc_hd__nand2_1)
    1.29  0.00  48.83 ^ core/U684/B2 (sky130_fd_sc_hd__o22ai_1)
1  0.00  0.33  0.64  49.47 v core/U684/Y (sky130_fd_sc_hd__o22ai_1)
    0.33  0.00  49.47 v core/CPU_Xreg_value_a4_reg[27][30]/D (sky130_fd_sc_hd__dfxtp_1)
    49.47 data arrival time

0.00 11.00 11.00 clock clk (rise edge)
    0.00 11.00 clock network delay (ideal)
    0.00 11.00 clock reconvergence pessimism
    11.00 ^ core/CPU_Xreg_value_a4_reg[27][30]/CLK (sky130_fd_sc_hd__dfxtp_1)
-1.23 9.77 library setup time
    9.77 data required time

-----

    9.77 data required time
-49.47 data arrival time

```

-39.70 slack (VIOLATED)

----- WNS -----

wns max -39.70

----- TNS -----

tns max -7993.79

=====
END OF LIBRARY: sky130_fd_sc_hd__ss_n40C_1v35.lib
=====

=====
LIBRARY: sky130_fd_sc_hd__ss_n40C_1v40.lib
=====

Warning 1140:
/data/Desktop/work/tools/openlane_working_dir/OpenSTA/vsdbabysoc/lib/sky130_fd_sc_hd__ss_n
40C_1v40.lib line 1, library sky130_fd_sc_hd__ss_n40C_1v40 already exists.

Warning 1140:
/data/Desktop/work/tools/openlane_working_dir/OpenSTA/vsdbabysoc/lib/avsddac.lib line 1, library
avsddac already exists.

Warning 1140:
/data/Desktop/work/tools/openlane_working_dir/OpenSTA/vsdbabysoc/lib/avsddac.lib line 1, library
avsddac already exists.

Warning 1140:
/data/Desktop/work/tools/openlane_working_dir/OpenSTA/vsdbabysoc/lib/avsdpll.lib line 1, library
avsdpll already exists.

Warning 1140:
/data/Desktop/work/tools/openlane_working_dir/OpenSTA/vsdbabysoc/lib/avsdpll.lib line 1, library
avsdpll already exists.

----- REPORT CHECKS -----

Warning 168: run_final.tcl line 1, unknown field nets.

Startpoint: core/CPU_imm_a2_reg[10]

(rising edge-triggered flip-flop clocked by clk)

Endpoint: core/CPU_imm_a3_reg[10]

(rising edge-triggered flip-flop clocked by clk)

Path Group: clk

Path Type: min

Fanout	Cap	Slew	Delay	Time	Description
--------	-----	------	-------	------	-------------

	0.00	0.00	0.00		clock clk (rise edge)
		0.00	0.00		clock network delay (ideal)
	0.00	0.00	0.00		^ core/CPU_imm_a2_reg[10]/CLK (sky130_fd_sc_hd__dfxtp_1)
1	0.00	0.05	0.91	0.91	v core/CPU_imm_a2_reg[10]/Q (sky130_fd_sc_hd__dfxtp_1)
	0.05	0.00	0.91		v core/CPU_imm_a3_reg[10]/D (sky130_fd_sc_hd__dfxtp_1)
			0.91		data arrival time
	0.00	0.00	0.00		clock clk (rise edge)
		0.00	0.00		clock network delay (ideal)
	0.00	0.00			clock reconvergence pessimism
		0.00			^ core/CPU_imm_a3_reg[10]/CLK (sky130_fd_sc_hd__dfxtp_1)

-0.22 -0.22 library hold time

-0.22 data required time

-0.22 data required time

-0.91 data arrival time

1.12 slack (MET)

Startpoint: core/CPU_is_addi_a3_reg

(rising edge-triggered flip-flop clocked by clk)

Endpoint: core/CPU_Xreg_value_a4_reg[27][30]

(rising edge-triggered flip-flop clocked by clk)

Path Group: clk

Path Type: max

Fanout	Cap	Slew	Delay	Time	Description
--------	-----	------	-------	------	-------------

	0.00	0.00	0.00		clock clk (rise edge)
	0.00	0.00			clock network delay (ideal)
	0.00	0.00	0.00		^ core/CPU_is_addi_a3_reg/CLK (sky130_fd_sc_hd__dfxtp_1)
2	0.00	0.08	0.94	0.94	v core/CPU_is_addi_a3_reg/Q (sky130_fd_sc_hd__dfxtp_1)
	0.08	0.00	0.94		v core/U474/A (sky130_fd_sc_hd__nor2_1)
2	0.00	0.39	0.43	1.38	^ core/U474/Y (sky130_fd_sc_hd__nor2_1)
	0.39	0.00	1.38		^ core/U476/A (sky130_fd_sc_hd__nand2_1)
2	0.01	0.17	0.34	1.72	v core/U476/Y (sky130_fd_sc_hd__nand2_1)
	0.17	0.00	1.72		v core/U9/A (sky130_fd_sc_hd__inv_2)

34 0.14 2.13 1.70 3.42 ^ core/U9/Y (sky130_fd_sc_hd__inv_2)
 2.13 0.00 3.42 ^ core/U479/B (sky130_fd_sc_hd__xor2_1)
 2 0.01 0.36 0.99 4.40 v core/U479/X (sky130_fd_sc_hd__xor2_1)
 0.36 0.00 4.40 v core/U569/A2 (sky130_fd_sc_hd__a21oi_1)
 2 0.01 0.64 0.87 5.27 ^ core/U569/Y (sky130_fd_sc_hd__a21oi_1)
 0.64 0.00 5.27 ^ core/U571/A2 (sky130_fd_sc_hd__o21ai_1)
 2 0.01 0.15 0.44 5.71 v core/U571/Y (sky130_fd_sc_hd__o21ai_1)
 0.15 0.00 5.71 v core/U574/A1 (sky130_fd_sc_hd__a21oi_1)
 2 0.01 0.64 0.71 6.43 ^ core/U574/Y (sky130_fd_sc_hd__a21oi_1)
 0.64 0.00 6.43 ^ core/U576/A2 (sky130_fd_sc_hd__o21ai_1)
 2 0.01 0.15 0.44 6.87 v core/U576/Y (sky130_fd_sc_hd__o21ai_1)
 0.15 0.00 6.87 v core/U581/A1 (sky130_fd_sc_hd__a21oi_1)
 2 0.01 0.64 0.71 7.58 ^ core/U581/Y (sky130_fd_sc_hd__a21oi_1)
 0.64 0.00 7.58 ^ core/U583/A2 (sky130_fd_sc_hd__o21ai_1)
 2 0.01 0.15 0.44 8.02 v core/U583/Y (sky130_fd_sc_hd__o21ai_1)
 0.15 0.00 8.02 v core/U587/A1 (sky130_fd_sc_hd__a21oi_1)
 2 0.01 0.64 0.71 8.74 ^ core/U587/Y (sky130_fd_sc_hd__a21oi_1)
 0.64 0.00 8.74 ^ core/U589/A2 (sky130_fd_sc_hd__o21ai_1)
 2 0.01 0.15 0.44 9.18 v core/U589/Y (sky130_fd_sc_hd__o21ai_1)
 0.15 0.00 9.18 v core/U344/A1 (sky130_fd_sc_hd__a21oi_1)
 2 0.01 0.64 0.71 9.89 ^ core/U344/Y (sky130_fd_sc_hd__a21oi_1)
 0.64 0.00 9.89 ^ core/U213/A2 (sky130_fd_sc_hd__o21ai_1)
 2 0.01 0.15 0.44 10.33 v core/U213/Y (sky130_fd_sc_hd__o21ai_1)
 0.15 0.00 10.33 v core/U343/A1 (sky130_fd_sc_hd__a21oi_1)
 2 0.01 0.64 0.71 11.05 ^ core/U343/Y (sky130_fd_sc_hd__a21oi_1)
 0.64 0.00 11.05 ^ core/U210/A2 (sky130_fd_sc_hd__o21ai_1)
 2 0.01 0.15 0.44 11.49 v core/U210/Y (sky130_fd_sc_hd__o21ai_1)

0.15 0.00 11.49 v core/U342/A1 (sky130_fd_sc_hd__a21oi_1)

2 0.01 0.64 0.71 12.20 ^ core/U342/Y (sky130_fd_sc_hd__a21oi_1)

0.64 0.00 12.20 ^ core/U209/A2 (sky130_fd_sc_hd__o21ai_1)

2 0.01 0.15 0.44 12.64 v core/U209/Y (sky130_fd_sc_hd__o21ai_1)

0.15 0.00 12.64 v core/U341/A1 (sky130_fd_sc_hd__a21oi_1)

2 0.01 0.64 0.71 13.36 ^ core/U341/Y (sky130_fd_sc_hd__a21oi_1)

0.64 0.00 13.36 ^ core/U215/A2 (sky130_fd_sc_hd__o21ai_1)

2 0.01 0.15 0.44 13.80 v core/U215/Y (sky130_fd_sc_hd__o21ai_1)

0.15 0.00 13.80 v core/U96/A1 (sky130_fd_sc_hd__a21o_1)

1 0.00 0.10 0.66 14.46 v core/U96/X (sky130_fd_sc_hd__a21o_1)

0.10 0.00 14.46 v core/U942/CIN (sky130_fd_sc_hd__fa_1)

2 0.01 0.28 1.95 16.41 v core/U942/COUT (sky130_fd_sc_hd__fa_1)

0.28 0.00 16.41 v core/U94/A1 (sky130_fd_sc_hd__a21o_1)

1 0.00 0.10 0.74 17.14 v core/U94/X (sky130_fd_sc_hd__a21o_1)

0.10 0.00 17.14 v core/U905/CIN (sky130_fd_sc_hd__fa_1)

1 0.00 0.25 1.91 19.05 v core/U905/COUT (sky130_fd_sc_hd__fa_1)

0.25 0.00 19.05 v core/U887/CIN (sky130_fd_sc_hd__fa_1)

1 0.00 0.25 1.98 21.04 v core/U887/COUT (sky130_fd_sc_hd__fa_1)

0.25 0.00 21.04 v core/U1370/CIN (sky130_fd_sc_hd__fa_1)

1 0.00 0.26 2.00 23.03 v core/U1370/COUT (sky130_fd_sc_hd__fa_1)

0.26 0.00 23.03 v core/U36/CIN (sky130_fd_sc_hd__fa_2)

2 0.01 0.24 1.79 24.82 v core/U36/COUT (sky130_fd_sc_hd__fa_2)

0.24 0.00 24.82 v core/U87/A (sky130_fd_sc_hd__clkinv_1)

1 0.00 0.08 0.22 25.04 ^ core/U87/Y (sky130_fd_sc_hd__clkinv_1)

0.08 0.00 25.04 ^ core/U613/A2 (sky130_fd_sc_hd__o21ai_1)

1 0.00 0.12 0.14 25.18 v core/U613/Y (sky130_fd_sc_hd__o21ai_1)

0.12 0.00 25.18 v core/U352/CIN (sky130_fd_sc_hd__fa_1)

```

1  0.00  0.25  1.92  27.10 v core/U352/COUT (sky130_fd_sc_hd__fa_1)
    0.25  0.00  27.10 v core/U348/CIN (sky130_fd_sc_hd__fa_1)
1  0.00  0.25  1.98  29.09 v core/U348/COUT (sky130_fd_sc_hd__fa_1)
    0.25  0.00  29.09 v core/U35/CIN (sky130_fd_sc_hd__fa_1)
1  0.00  0.25  1.98  31.07 v core/U35/COUT (sky130_fd_sc_hd__fa_1)
    0.25  0.00  31.07 v core/U33/CIN (sky130_fd_sc_hd__fa_1)
2  0.01  0.28  2.02  33.10 v core/U33/COUT (sky130_fd_sc_hd__fa_1)
    0.28  0.00  33.10 v core/U81/A1 (sky130_fd_sc_hd__a21o_1)
1  0.00  0.10  0.74  33.83 v core/U81/X (sky130_fd_sc_hd__a21o_1)
    0.10  0.00  33.83 v core/U347/CIN (sky130_fd_sc_hd__fa_1)
1  0.00  0.25  1.91  35.74 v core/U347/COUT (sky130_fd_sc_hd__fa_1)
    0.25  0.00  35.74 v core/U351/CIN (sky130_fd_sc_hd__fa_1)
2  0.00  0.28  2.46  38.20 v core/U351/SUM (sky130_fd_sc_hd__fa_1)
    0.28  0.00  38.20 v core/U3/A (sky130_fd_sc_hd__nand2_1)
15 0.03  1.09  1.00  39.20 ^ core/U3/Y (sky130_fd_sc_hd__nand2_1)
    1.09  0.00  39.20 ^ core/U684/B2 (sky130_fd_sc_hd__o22ai_1)
1  0.00  0.29  0.53  39.74 v core/U684/Y (sky130_fd_sc_hd__o22ai_1)
    0.29  0.00  39.74 v core/CPU_Xreg_value_a4_reg[27][30]/D (sky130_fd_sc_hd__dfxtp_1)
    39.74  data arrival time

0.00 11.00 11.00 clock clk (rise edge)
    0.00 11.00 clock network delay (ideal)
    0.00 11.00 clock reconvergence pessimism
    11.00 ^ core/CPU_Xreg_value_a4_reg[27][30]/CLK (sky130_fd_sc_hd__dfxtp_1)
-0.96 10.04 library setup time
    10.04 data required time

```

10.04 data required time

-39.74 data arrival time

-29.69 slack (VIOLATED)

----- WNS -----

wns max -29.69

----- TNS -----

tns max -5466.48

=====

END OF LIBRARY: sky130_fd_sc_hd__ss_n40C_1v40.lib

=====

=====

LIBRARY: sky130_fd_sc_hd__ss_n40C_1v44.lib

=====

Warning 1140:
/data/Desktop/work/tools/openlane_working_dir/OpenSTA/vsdbabysoc/lib/sky130_fd_sc_hd__ss_n40C_1v44.lib line 1, library sky130_fd_sc_hd__ss_n40C_1v44 already exists.

Warning 1140:
/data/Desktop/work/tools/openlane_working_dir/OpenSTA/vsdbabysoc/lib/avsdac.lib line 1, library avsdac already exists.

Warning 1140:
/data/Desktop/work/tools/openlane_working_dir/OpenSTA/vsdbabysoc/lib/avsdac.lib line 1, library
avsdac already exists.

Warning 1140:
/data/Desktop/work/tools/openlane_working_dir/OpenSTA/vsdbabysoc/lib/avsdpll.lib line 1, library
avsdpll already exists.

Warning 1140:
/data/Desktop/work/tools/openlane_working_dir/OpenSTA/vsdbabysoc/lib/avsdpll.lib line 1, library
avsdpll already exists.

----- REPORT CHECKS -----

Warning 168: run_final.tcl line 1, unknown field nets.

Startpoint: core/CPU_imm_a2_reg[10]

(rising edge-triggered flip-flop clocked by clk)

Endpoint: core/CPU_imm_a3_reg[10]

(rising edge-triggered flip-flop clocked by clk)

Path Group: clk

Path Type: min

Fanout	Cap	Slew	Delay	Time	Description
--------	-----	------	-------	------	-------------

0.00	0.00	0.00	clock	clk (rise edge)
------	------	------	-------	-----------------

0.00	0.00	clock	network delay (ideal)
------	------	-------	-----------------------

0.00	0.00	0.00	^	core/CPU_imm_a2_reg[10]/CLK (sky130_fd_sc_hd__dfxtp_1)
------	------	------	---	--

1	0.00	0.05	0.80	0.80	v core/CPU_imm_a2_reg[10]/Q (sky130_fd_sc_hd__dfxtp_1)
---	------	------	------	------	--

0.05	0.00	0.80	v	core/CPU_imm_a3_reg[10]/D (sky130_fd_sc_hd__dfxtp_1)
------	------	------	---	--

0.80	data	arrival	time
------	------	---------	------

0.00	0.00	0.00	clock	clk (rise edge)
------	------	------	-------	-----------------

0.00	0.00	clock	network delay (ideal)
------	------	-------	-----------------------

0.00	0.00	clock reconvergence pessimism
0.00	^ core/CPU_imm_a3_reg[10]/CLK (sky130_fd_sc_hd__dfxtp_1)	
-0.19	-0.19	library hold time
-0.19	data required time	

-0.19	data required time	
-0.80	data arrival time	

0.99	slack (MET)	

Startpoint: core/CPU_is_addi_a3_reg
(rising edge-triggered flip-flop clocked by clk)
Endpoint: core/CPU_Xreg_value_a4_reg[27][30]
(rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max

Fanout	Cap	Slew	Delay	Time	Description

0.00	0.00	0.00	clock clk (rise edge)		
0.00	0.00	clock network delay (ideal)			
0.00	0.00	0.00	^ core/CPU_is_addi_a3_reg/CLK (sky130_fd_sc_hd__dfxtp_1)		
2	0.00	0.07	0.83	0.83	v core/CPU_is_addi_a3_reg/Q (sky130_fd_sc_hd__dfxtp_1)
	0.07	0.00	0.83	v core/U474/A (sky130_fd_sc_hd__nor2_1)	
2	0.00	0.34	0.38	1.21	^ core/U474/Y (sky130_fd_sc_hd__nor2_1)
	0.34	0.00	1.21	^ core/U476/A (sky130_fd_sc_hd__nand2_1)	

2 0.01 0.15 0.30 1.51 v core/U476/Y (sky130_fd_sc_hd__nand2_1)
 0.15 0.00 1.51 v core/U9/A (sky130_fd_sc_hd__inv_2)
 34 0.14 1.89 1.50 3.01 ^ core/U9/Y (sky130_fd_sc_hd__inv_2)
 1.89 0.00 3.01 ^ core/U479/B (sky130_fd_sc_hd__xor2_1)
 2 0.01 0.34 0.88 3.88 v core/U479/X (sky130_fd_sc_hd__xor2_1)
 0.34 0.00 3.88 v core/U569/A2 (sky130_fd_sc_hd__a21oi_1)
 2 0.01 0.56 0.77 4.65 ^ core/U569/Y (sky130_fd_sc_hd__a21oi_1)
 0.56 0.00 4.65 ^ core/U571/A2 (sky130_fd_sc_hd__o21ai_1)
 2 0.01 0.14 0.39 5.04 v core/U571/Y (sky130_fd_sc_hd__o21ai_1)
 0.14 0.00 5.04 v core/U574/A1 (sky130_fd_sc_hd__a21oi_1)
 2 0.01 0.56 0.62 5.67 ^ core/U574/Y (sky130_fd_sc_hd__a21oi_1)
 0.56 0.00 5.67 ^ core/U576/A2 (sky130_fd_sc_hd__o21ai_1)
 2 0.01 0.14 0.39 6.06 v core/U576/Y (sky130_fd_sc_hd__o21ai_1)
 0.14 0.00 6.06 v core/U581/A1 (sky130_fd_sc_hd__a21oi_1)
 2 0.01 0.56 0.62 6.68 ^ core/U581/Y (sky130_fd_sc_hd__a21oi_1)
 0.56 0.00 6.68 ^ core/U583/A2 (sky130_fd_sc_hd__o21ai_1)
 2 0.01 0.14 0.39 7.07 v core/U583/Y (sky130_fd_sc_hd__o21ai_1)
 0.14 0.00 7.07 v core/U587/A1 (sky130_fd_sc_hd__a21oi_1)
 2 0.01 0.56 0.62 7.69 ^ core/U587/Y (sky130_fd_sc_hd__a21oi_1)
 0.56 0.00 7.69 ^ core/U589/A2 (sky130_fd_sc_hd__o21ai_1)
 2 0.01 0.14 0.39 8.08 v core/U589/Y (sky130_fd_sc_hd__o21ai_1)
 0.14 0.00 8.08 v core/U344/A1 (sky130_fd_sc_hd__a21oi_1)
 2 0.01 0.56 0.62 8.71 ^ core/U344/Y (sky130_fd_sc_hd__a21oi_1)
 0.56 0.00 8.71 ^ core/U213/A2 (sky130_fd_sc_hd__o21ai_1)
 2 0.01 0.14 0.39 9.10 v core/U213/Y (sky130_fd_sc_hd__o21ai_1)
 0.14 0.00 9.10 v core/U343/A1 (sky130_fd_sc_hd__a21oi_1)
 2 0.01 0.56 0.62 9.72 ^ core/U343/Y (sky130_fd_sc_hd__a21oi_1)

0.56 0.00 9.72 ^ core/U210/A2 (sky130_fd_sc_hd__o21ai_1)

2 0.01 0.14 0.39 10.11 v core/U210/Y (sky130_fd_sc_hd__o21ai_1)

0.14 0.00 10.11 v core/U342/A1 (sky130_fd_sc_hd__a21oi_1)

2 0.01 0.56 0.62 10.73 ^ core/U342/Y (sky130_fd_sc_hd__a21oi_1)

0.56 0.00 10.73 ^ core/U209/A2 (sky130_fd_sc_hd__o21ai_1)

2 0.01 0.14 0.39 11.12 v core/U209/Y (sky130_fd_sc_hd__o21ai_1)

0.14 0.00 11.12 v core/U341/A1 (sky130_fd_sc_hd__a21oi_1)

2 0.01 0.56 0.62 11.75 ^ core/U341/Y (sky130_fd_sc_hd__a21oi_1)

0.56 0.00 11.75 ^ core/U215/A2 (sky130_fd_sc_hd__o21ai_1)

2 0.01 0.14 0.39 12.14 v core/U215/Y (sky130_fd_sc_hd__o21ai_1)

0.14 0.00 12.14 v core/U96/A1 (sky130_fd_sc_hd__a21o_1)

1 0.00 0.09 0.57 12.71 v core/U96/X (sky130_fd_sc_hd__a21o_1)

0.09 0.00 12.71 v core/U942/CIN (sky130_fd_sc_hd__fa_1)

2 0.01 0.24 1.65 14.36 v core/U942/COUT (sky130_fd_sc_hd__fa_1)

0.24 0.00 14.36 v core/U94/A1 (sky130_fd_sc_hd__a21o_1)

1 0.00 0.09 0.63 15.00 v core/U94/X (sky130_fd_sc_hd__a21o_1)

0.09 0.00 15.00 v core/U905/CIN (sky130_fd_sc_hd__fa_1)

1 0.00 0.22 1.61 16.61 v core/U905/COUT (sky130_fd_sc_hd__fa_1)

0.22 0.00 16.61 v core/U887/CIN (sky130_fd_sc_hd__fa_1)

1 0.00 0.22 1.67 18.28 v core/U887/COUT (sky130_fd_sc_hd__fa_1)

0.22 0.00 18.28 v core/U1370/CIN (sky130_fd_sc_hd__fa_1)

1 0.00 0.22 1.68 19.96 v core/U1370/COUT (sky130_fd_sc_hd__fa_1)

0.22 0.00 19.96 v core/U36/CIN (sky130_fd_sc_hd__fa_2)

2 0.01 0.20 1.51 21.48 v core/U36/COUT (sky130_fd_sc_hd__fa_2)

0.20 0.00 21.48 v core/U87/A (sky130_fd_sc_hd__clkinv_1)

1 0.00 0.07 0.19 21.67 ^ core/U87/Y (sky130_fd_sc_hd__clkinv_1)

0.07 0.00 21.67 ^ core/U613/A2 (sky130_fd_sc_hd__o21ai_1)

```

1  0.00  0.11  0.13  21.79 v core/U613/Y (sky130_fd_sc_hd__o21ai_1)
    0.11  0.00  21.79 v core/U352/CIN (sky130_fd_sc_hd__fa_1)
1  0.00  0.22  1.62  23.42 v core/U352/COUT (sky130_fd_sc_hd__fa_1)
    0.22  0.00  23.42 v core/U348/CIN (sky130_fd_sc_hd__fa_1)
1  0.00  0.22  1.67  25.09 v core/U348/COUT (sky130_fd_sc_hd__fa_1)
    0.22  0.00  25.09 v core/U35/CIN (sky130_fd_sc_hd__fa_1)
1  0.00  0.22  1.67  26.76 v core/U35/COUT (sky130_fd_sc_hd__fa_1)
    0.22  0.00  26.76 v core/U33/CIN (sky130_fd_sc_hd__fa_1)
2  0.01  0.24  1.71  28.47 v core/U33/COUT (sky130_fd_sc_hd__fa_1)
    0.24  0.00  28.47 v core/U81/A1 (sky130_fd_sc_hd__a21o_1)
1  0.00  0.09  0.63  29.10 v core/U81/X (sky130_fd_sc_hd__a21o_1)
    0.09  0.00  29.10 v core/U347/CIN (sky130_fd_sc_hd__fa_1)
1  0.00  0.22  1.61  30.72 v core/U347/COUT (sky130_fd_sc_hd__fa_1)
    0.22  0.00  30.72 v core/U351/CIN (sky130_fd_sc_hd__fa_1)
2  0.00  0.24  2.07  32.79 v core/U351/SUM (sky130_fd_sc_hd__fa_1)
    0.24  0.00  32.79 v core/U3/A (sky130_fd_sc_hd__nand2_1)
15 0.03  0.96  0.87  33.65 ^ core/U3/Y (sky130_fd_sc_hd__nand2_1)
    0.96  0.00  33.65 ^ core/U684/B2 (sky130_fd_sc_hd__o22ai_1)
1  0.00  0.26  0.45  34.10 v core/U684/Y (sky130_fd_sc_hd__o22ai_1)
    0.26  0.00  34.10 v core/CPU_Xreg_value_a4_reg[27][30]/D (sky130_fd_sc_hd__dfxtp_1)
    34.10 data arrival time

0.00 11.00 11.00 clock clk (rise edge)
    0.00 11.00 clock network delay (ideal)
    0.00 11.00 clock reconvergence pessimism
    11.00 ^ core/CPU_Xreg_value_a4_reg[27][30]/CLK (sky130_fd_sc_hd__dfxtp_1)
    -0.80 10.20 library setup time

```

10.20 data required time

10.20 data required time

-34.10 data arrival time

-23.91 slack (VIOLATED)

----- WNS -----

wns max -23.91

----- TNS -----

tns max -4107.19

=====
END OF LIBRARY: sky130_fd_sc_hd__ss_n40C_1v44.lib
=====

=====
LIBRARY: sky130_fd_sc_hd__ss_n40C_1v60.lib
=====

Warning 1140:
/data/Desktop/work/tools/openlane_working_dir/OpenSTA/vsdbabysoc/lib/sky130_fd_sc_hd__ss_n
40C_1v60.lib line 1, library sky130_fd_sc_hd__ss_n40C_1v60 already exists.

Warning 1140:
/data/Desktop/work/tools/openlane_working_dir/OpenSTA/vsdbabysoc/lib/avsddac.lib line 1, library
avsddac already exists.

Warning 1140:
/data/Desktop/work/tools/openlane_working_dir/OpenSTA/vsdbabysoc/lib/avsddac.lib line 1, library
avsddac already exists.

Warning 1140:
/data/Desktop/work/tools/openlane_working_dir/OpenSTA/vsdbabysoc/lib/avsdpll.lib line 1, library
avsdpll already exists.

Warning 1140:
/data/Desktop/work/tools/openlane_working_dir/OpenSTA/vsdbabysoc/lib/avsdpll.lib line 1, library
avsdpll already exists.

----- REPORT CHECKS -----

Warning 168: run_final.tcl line 1, unknown field nets.

Startpoint: core/CPU_imm_a2_reg[10]

(rising edge-triggered flip-flop clocked by clk)

Endpoint: core/CPU_imm_a3_reg[10]

(rising edge-triggered flip-flop clocked by clk)

Path Group: clk

Path Type: min

Fanout	Cap	Slew	Delay	Time	Description
--------	-----	------	-------	------	-------------

0.00	0.00	0.00	clock clk (rise edge)		
------	------	------	-----------------------	--	--

0.00	0.00	clock network delay (ideal)		
------	------	-----------------------------	--	--

0.00	0.00	0.00	^ core/CPU_imm_a2_reg[10]/CLK (sky130_fd_sc_hd__dfxtp_1)		
------	------	------	--	--	--

1	0.00	0.03	0.54	0.54	v core/CPU_imm_a2_reg[10]/Q (sky130_fd_sc_hd__dfxtp_1)
---	------	------	------	------	--

0.03	0.00	0.54	v core/CPU_imm_a3_reg[10]/D (sky130_fd_sc_hd__dfxtp_1)		
------	------	------	--	--	--

0.54	data arrival time				
------	-------------------	--	--	--	--

	0.00	0.00	0.00		clock clk (rise edge)
		0.00	0.00		clock network delay (ideal)
		0.00	0.00		clock reconvergence pessimism
			0.00	^	core/CPU_imm_a3_reg[10]/CLK (sky130_fd_sc_hd__dfxtp_1)
	-0.12	-0.12			library hold time
		-0.12			data required time

		-0.12			data required time
		-0.54			data arrival time

		0.66			slack (MET)
Startpoint: core/CPU_is_addi_a3_reg					
(rising edge-triggered flip-flop clocked by clk)					
Endpoint: core/CPU_Xreg_value_a4_reg[27][30]					
(rising edge-triggered flip-flop clocked by clk)					
Path Group: clk					
Path Type: max					
Fanout	Cap	Slew	Delay	Time	Description

	0.00	0.00	0.00		clock clk (rise edge)
		0.00	0.00		clock network delay (ideal)
	0.00	0.00	0.00	^	core/CPU_is_addi_a3_reg/CLK (sky130_fd_sc_hd__dfxtp_1)
2	0.00	0.05	0.57	0.57	v core/CPU_is_addi_a3_reg/Q (sky130_fd_sc_hd__dfxtp_1)
	0.05	0.00	0.57	v	core/U474/A (sky130_fd_sc_hd__nor2_1)

2 0.00 0.23 0.25 0.82 ^ core/U474/Y (sky130_fd_sc_hd__nor2_1)
 0.23 0.00 0.82 ^ core/U476/A (sky130_fd_sc_hd__nand2_1)
 2 0.01 0.12 0.20 1.02 v core/U476/Y (sky130_fd_sc_hd__nand2_1)
 0.12 0.00 1.02 v core/U9/A (sky130_fd_sc_hd__inv_2)
 34 0.14 1.32 1.04 2.06 ^ core/U9/Y (sky130_fd_sc_hd__inv_2)
 1.32 0.00 2.06 ^ core/U479/B (sky130_fd_sc_hd__xor2_1)
 2 0.01 0.25 0.53 2.58 v core/U479/X (sky130_fd_sc_hd__xor2_1)
 0.25 0.00 2.58 v core/U569/A2 (sky130_fd_sc_hd__a21oi_1)
 2 0.01 0.38 0.52 3.10 ^ core/U569/Y (sky130_fd_sc_hd__a21oi_1)
 0.38 0.00 3.10 ^ core/U571/A2 (sky130_fd_sc_hd__o21ai_1)
 2 0.01 0.11 0.24 3.35 v core/U571/Y (sky130_fd_sc_hd__o21ai_1)
 0.11 0.00 3.35 v core/U574/A1 (sky130_fd_sc_hd__a21oi_1)
 2 0.01 0.38 0.41 3.76 ^ core/U574/Y (sky130_fd_sc_hd__a21oi_1)
 0.38 0.00 3.76 ^ core/U576/A2 (sky130_fd_sc_hd__o21ai_1)
 2 0.01 0.11 0.24 4.00 v core/U576/Y (sky130_fd_sc_hd__o21ai_1)
 0.11 0.00 4.00 v core/U581/A1 (sky130_fd_sc_hd__a21oi_1)
 2 0.01 0.38 0.41 4.41 ^ core/U581/Y (sky130_fd_sc_hd__a21oi_1)
 0.38 0.00 4.41 ^ core/U583/A2 (sky130_fd_sc_hd__o21ai_1)
 2 0.01 0.11 0.24 4.66 v core/U583/Y (sky130_fd_sc_hd__o21ai_1)
 0.11 0.00 4.66 v core/U587/A1 (sky130_fd_sc_hd__a21oi_1)
 2 0.01 0.38 0.41 5.06 ^ core/U587/Y (sky130_fd_sc_hd__a21oi_1)
 0.38 0.00 5.06 ^ core/U589/A2 (sky130_fd_sc_hd__o21ai_1)
 2 0.01 0.11 0.24 5.31 v core/U589/Y (sky130_fd_sc_hd__o21ai_1)
 0.11 0.00 5.31 v core/U344/A1 (sky130_fd_sc_hd__a21oi_1)
 2 0.01 0.38 0.41 5.72 ^ core/U344/Y (sky130_fd_sc_hd__a21oi_1)
 0.38 0.00 5.72 ^ core/U213/A2 (sky130_fd_sc_hd__o21ai_1)
 2 0.01 0.11 0.24 5.96 v core/U213/Y (sky130_fd_sc_hd__o21ai_1)

0.11 0.00 5.96 v core/U343/A1 (sky130_fd_sc_hd__a21oi_1)

2 0.01 0.38 0.41 6.37 ^ core/U343/Y (sky130_fd_sc_hd__a21oi_1)

0.38 0.00 6.37 ^ core/U210/A2 (sky130_fd_sc_hd__o21ai_1)

2 0.01 0.11 0.24 6.62 v core/U210/Y (sky130_fd_sc_hd__o21ai_1)

0.11 0.00 6.62 v core/U342/A1 (sky130_fd_sc_hd__a21oi_1)

2 0.01 0.38 0.41 7.03 ^ core/U342/Y (sky130_fd_sc_hd__a21oi_1)

0.38 0.00 7.03 ^ core/U209/A2 (sky130_fd_sc_hd__o21ai_1)

2 0.01 0.11 0.24 7.27 v core/U209/Y (sky130_fd_sc_hd__o21ai_1)

0.11 0.00 7.27 v core/U341/A1 (sky130_fd_sc_hd__a21oi_1)

2 0.01 0.38 0.41 7.68 ^ core/U341/Y (sky130_fd_sc_hd__a21oi_1)

0.38 0.00 7.68 ^ core/U215/A2 (sky130_fd_sc_hd__o21ai_1)

2 0.01 0.11 0.25 7.93 v core/U215/Y (sky130_fd_sc_hd__o21ai_1)

0.11 0.00 7.93 v core/U96/A1 (sky130_fd_sc_hd__a21o_1)

1 0.00 0.06 0.37 8.30 v core/U96/X (sky130_fd_sc_hd__a21o_1)

0.06 0.00 8.30 v core/U942/CIN (sky130_fd_sc_hd__fa_1)

2 0.01 0.16 0.98 9.28 v core/U942/COUT (sky130_fd_sc_hd__fa_1)

0.16 0.00 9.28 v core/U94/A1 (sky130_fd_sc_hd__a21o_1)

1 0.00 0.06 0.40 9.68 v core/U94/X (sky130_fd_sc_hd__a21o_1)

0.06 0.00 9.68 v core/U905/CIN (sky130_fd_sc_hd__fa_1)

1 0.00 0.14 0.96 10.64 v core/U905/COUT (sky130_fd_sc_hd__fa_1)

0.14 0.00 10.64 v core/U887/CIN (sky130_fd_sc_hd__fa_1)

1 0.00 0.14 0.99 11.63 v core/U887/COUT (sky130_fd_sc_hd__fa_1)

0.14 0.00 11.63 v core/U1370/CIN (sky130_fd_sc_hd__fa_1)

1 0.00 0.15 1.00 12.63 v core/U1370/COUT (sky130_fd_sc_hd__fa_1)

0.15 0.00 12.63 v core/U36/CIN (sky130_fd_sc_hd__fa_2)

2 0.01 0.13 0.91 13.53 v core/U36/COUT (sky130_fd_sc_hd__fa_2)

0.13 0.00 13.53 v core/U87/A (sky130_fd_sc_hd__clkinv_1)

```

1  0.00  0.05  0.12  13.65 ^ core/U87/Y (sky130_fd_sc_hd__clkinv_1)
    0.05  0.00  13.65 ^ core/U613/A2 (sky130_fd_sc_hd__o21ai_1)
1  0.00  0.08  0.09  13.74 v core/U613/Y (sky130_fd_sc_hd__o21ai_1)
    0.08  0.00  13.74 v core/U352/CIN (sky130_fd_sc_hd__fa_1)
1  0.00  0.14  0.96  14.71 v core/U352/COUT (sky130_fd_sc_hd__fa_1)
    0.14  0.00  14.71 v core/U348/CIN (sky130_fd_sc_hd__fa_1)
1  0.00  0.14  0.99  15.70 v core/U348/COUT (sky130_fd_sc_hd__fa_1)
    0.14  0.00  15.70 v core/U35/CIN (sky130_fd_sc_hd__fa_1)
1  0.00  0.14  0.99  16.69 v core/U35/COUT (sky130_fd_sc_hd__fa_1)
    0.14  0.00  16.69 v core/U33/CIN (sky130_fd_sc_hd__fa_1)
2  0.01  0.16  1.02  17.71 v core/U33/COUT (sky130_fd_sc_hd__fa_1)
    0.16  0.00  17.71 v core/U81/A1 (sky130_fd_sc_hd__a21o_1)
1  0.00  0.06  0.40  18.11 v core/U81/X (sky130_fd_sc_hd__a21o_1)
    0.06  0.00  18.11 v core/U347/CIN (sky130_fd_sc_hd__fa_1)
1  0.00  0.14  0.96  19.06 v core/U347/COUT (sky130_fd_sc_hd__fa_1)
    0.14  0.00  19.06 v core/U351/CIN (sky130_fd_sc_hd__fa_1)
2  0.00  0.16  1.21  20.27 v core/U351/SUM (sky130_fd_sc_hd__fa_1)
    0.16  0.00  20.27 v core/U3/A (sky130_fd_sc_hd__nand2_1)
15 0.04  0.68  0.59  20.87 ^ core/U3/Y (sky130_fd_sc_hd__nand2_1)
    0.68  0.00  20.87 ^ core/U684/B2 (sky130_fd_sc_hd__o22ai_1)
1  0.00  0.20  0.29  21.15 v core/U684/Y (sky130_fd_sc_hd__o22ai_1)
    0.20  0.00  21.15 v core/CPU_Xreg_value_a4_reg[27][30]/D (sky130_fd_sc_hd__dfxtp_1)
    21.15 data arrival time

0.00 11.00 11.00 clock clk (rise edge)
    0.00 11.00 clock network delay (ideal)
    0.00 11.00 clock reconvergence pessimism

```

11.00 ^ core/CPU_Xreg_value_a4_reg[27][30]/CLK (sky130_fd_sc_hd__dfxtp_1)

-0.46 10.54 library setup time

10.54 data required time

10.54 data required time

-21.15 data arrival time

-10.61 slack (VIOLATED)

----- WNS -----

wns max -10.61

----- TNS -----

tns max -1355.80

=====

END OF LIBRARY: sky130_fd_sc_hd__ss_n40C_1v60.lib

=====

=====

LIBRARY: sky130_fd_sc_hd__ss_n40C_1v76.lib

=====

Warning 1140:
/data/Desktop/work/tools/openlane_working_dir/OpenSTA/vsdbabysoc/lib/sky130_fd_sc_hd__ss_n40C_1v76.lib line 1, library sky130_fd_sc_hd__ss_n40C_1v76 already exists.

Warning 1140:
/data/Desktop/work/tools/openlane_working_dir/OpenSTA/vsdbabysoc/lib/avsddac.lib line 1, library avsddac already exists.

Warning 1140:
/data/Desktop/work/tools/openlane_working_dir/OpenSTA/vsdbabysoc/lib/avsddac.lib line 1, library avsddac already exists.

Warning 1140:
/data/Desktop/work/tools/openlane_working_dir/OpenSTA/vsdbabysoc/lib/avsdpll.lib line 1, library avsdpll already exists.

Warning 1140:
/data/Desktop/work/tools/openlane_working_dir/OpenSTA/vsdbabysoc/lib/avsdpll.lib line 1, library avsdpll already exists.

----- REPORT CHECKS -----

Warning 168: run_final.tcl line 1, unknown field nets.

Startpoint: core/CPU_imm_a2_reg[10]

(rising edge-triggered flip-flop clocked by clk)

Endpoint: core/CPU_imm_a3_reg[10]

(rising edge-triggered flip-flop clocked by clk)

Path Group: clk

Path Type: min

Fanout	Cap	Slew	Delay	Time	Description
--------	-----	------	-------	------	-------------

0.00	0.00	0.00	clock clk (rise edge)		
------	------	------	-----------------------	--	--

0.00	0.00	clock network delay (ideal)		
------	------	-----------------------------	--	--

0.00	0.00	0.00	^ core/CPU_imm_a2_reg[10]/CLK (sky130_fd_sc_hd__dfxtp_1)		
------	------	------	--	--	--

1	0.00	0.03	0.41	0.41	v core/CPU_imm_a2_reg[10]/Q (sky130_fd_sc_hd__dfxtp_1)
---	------	------	------	------	--

0.03 0.00 0.41 v core/CPU_imm_a3_reg[10]/D (sky130_fd_sc_hd__dfxtp_1)

0.41 data arrival time

0.00 0.00 0.00 clock clk (rise edge)

0.00 0.00 clock network delay (ideal)

0.00 0.00 clock reconvergence pessimism

0.00 ^ core/CPU_imm_a3_reg[10]/CLK (sky130_fd_sc_hd__dfxtp_1)

-0.09 -0.09 library hold time

-0.09 data required time

-0.09 data required time

-0.41 data arrival time

0.50 slack (MET)

Startpoint: core/CPU_is_addi_a3_reg

(rising edge-triggered flip-flop clocked by clk)

Endpoint: core/CPU_Xreg_value_a4_reg[27][30]

(rising edge-triggered flip-flop clocked by clk)

Path Group: clk

Path Type: max

Fanout	Cap	Slew	Delay	Time	Description
--------	-----	------	-------	------	-------------

0.00 0.00 0.00 clock clk (rise edge)

0.00 0.00 clock network delay (ideal)

```

0.00 0.00 0.00 ^ core/CPU_is_addi_a3_reg/CLK (sky130_fd_sc_hd__dfxtp_1)
2 0.00 0.05 0.43 0.43 v core/CPU_is_addi_a3_reg/Q (sky130_fd_sc_hd__dfxtp_1)
0.05 0.00 0.43 v core/U474/A (sky130_fd_sc_hd__nor2_1)
2 0.00 0.17 0.19 0.62 ^ core/U474/Y (sky130_fd_sc_hd__nor2_1)
0.17 0.00 0.62 ^ core/U476/A (sky130_fd_sc_hd__nand2_1)
2 0.01 0.10 0.15 0.77 v core/U476/Y (sky130_fd_sc_hd__nand2_1)
0.10 0.00 0.77 v core/U9/A (sky130_fd_sc_hd__inv_2)
34 0.15 1.02 0.80 1.57 ^ core/U9/Y (sky130_fd_sc_hd__inv_2)
1.02 0.00 1.57 ^ core/U479/B (sky130_fd_sc_hd__xor2_1)
2 0.01 0.21 0.36 1.93 v core/U479/X (sky130_fd_sc_hd__xor2_1)
0.21 0.00 1.93 v core/U569/A2 (sky130_fd_sc_hd__a21oi_1)
2 0.01 0.29 0.39 2.32 ^ core/U569/Y (sky130_fd_sc_hd__a21oi_1)
0.29 0.00 2.32 ^ core/U571/A2 (sky130_fd_sc_hd__o21ai_1)
2 0.01 0.09 0.18 2.50 v core/U571/Y (sky130_fd_sc_hd__o21ai_1)
0.09 0.00 2.50 v core/U574/A1 (sky130_fd_sc_hd__a21oi_1)
2 0.01 0.29 0.31 2.80 ^ core/U574/Y (sky130_fd_sc_hd__a21oi_1)
0.29 0.00 2.80 ^ core/U576/A2 (sky130_fd_sc_hd__o21ai_1)
2 0.01 0.09 0.18 2.98 v core/U576/Y (sky130_fd_sc_hd__o21ai_1)
0.09 0.00 2.98 v core/U581/A1 (sky130_fd_sc_hd__a21oi_1)
2 0.01 0.29 0.31 3.29 ^ core/U581/Y (sky130_fd_sc_hd__a21oi_1)
0.29 0.00 3.29 ^ core/U583/A2 (sky130_fd_sc_hd__o21ai_1)
2 0.01 0.09 0.18 3.47 v core/U583/Y (sky130_fd_sc_hd__o21ai_1)
0.09 0.00 3.47 v core/U587/A1 (sky130_fd_sc_hd__a21oi_1)
2 0.01 0.29 0.31 3.77 ^ core/U587/Y (sky130_fd_sc_hd__a21oi_1)
0.29 0.00 3.77 ^ core/U589/A2 (sky130_fd_sc_hd__o21ai_1)
2 0.01 0.09 0.18 3.95 v core/U589/Y (sky130_fd_sc_hd__o21ai_1)
0.09 0.00 3.95 v core/U344/A1 (sky130_fd_sc_hd__a21oi_1)

```

2 0.01 0.29 0.31 4.26 ^ core/U344/Y (sky130_fd_sc_hd__a21oi_1)
 0.29 0.00 4.26 ^ core/U213/A2 (sky130_fd_sc_hd__o21ai_1)
 2 0.01 0.09 0.18 4.44 v core/U213/Y (sky130_fd_sc_hd__o21ai_1)
 0.09 0.00 4.44 v core/U343/A1 (sky130_fd_sc_hd__a21oi_1)
 2 0.01 0.29 0.31 4.74 ^ core/U343/Y (sky130_fd_sc_hd__a21oi_1)
 0.29 0.00 4.74 ^ core/U210/A2 (sky130_fd_sc_hd__o21ai_1)
 2 0.01 0.09 0.18 4.92 v core/U210/Y (sky130_fd_sc_hd__o21ai_1)
 0.09 0.00 4.92 v core/U342/A1 (sky130_fd_sc_hd__a21oi_1)
 2 0.01 0.29 0.31 5.23 ^ core/U342/Y (sky130_fd_sc_hd__a21oi_1)
 0.29 0.00 5.23 ^ core/U209/A2 (sky130_fd_sc_hd__o21ai_1)
 2 0.01 0.09 0.18 5.41 v core/U209/Y (sky130_fd_sc_hd__o21ai_1)
 0.09 0.00 5.41 v core/U341/A1 (sky130_fd_sc_hd__a21oi_1)
 2 0.01 0.29 0.31 5.71 ^ core/U341/Y (sky130_fd_sc_hd__a21oi_1)
 0.29 0.00 5.71 ^ core/U215/A2 (sky130_fd_sc_hd__o21ai_1)
 2 0.01 0.09 0.18 5.89 v core/U215/Y (sky130_fd_sc_hd__o21ai_1)
 0.09 0.00 5.89 v core/U96/A1 (sky130_fd_sc_hd__a21o_1)
 1 0.00 0.05 0.27 6.16 v core/U96/X (sky130_fd_sc_hd__a21o_1)
 0.05 0.00 6.16 v core/U942/CIN (sky130_fd_sc_hd__fa_1)
 2 0.01 0.12 0.68 6.84 v core/U942/COUT (sky130_fd_sc_hd__fa_1)
 0.12 0.00 6.84 v core/U94/A1 (sky130_fd_sc_hd__a21o_1)
 1 0.00 0.05 0.29 7.13 v core/U94/X (sky130_fd_sc_hd__a21o_1)
 0.05 0.00 7.13 v core/U905/CIN (sky130_fd_sc_hd__fa_1)
 1 0.00 0.11 0.65 7.79 v core/U905/COUT (sky130_fd_sc_hd__fa_1)
 0.11 0.00 7.79 v core/U887/CIN (sky130_fd_sc_hd__fa_1)
 1 0.00 0.11 0.68 8.47 v core/U887/COUT (sky130_fd_sc_hd__fa_1)
 0.11 0.00 8.47 v core/U1370/CIN (sky130_fd_sc_hd__fa_1)
 1 0.00 0.11 0.68 9.15 v core/U1370/COUT (sky130_fd_sc_hd__fa_1)

0.11 0.00 9.15 v core/U36/CIN (sky130_fd_sc_hd__fa_2)

2 0.01 0.10 0.63 9.78 v core/U36/COUT (sky130_fd_sc_hd__fa_2)

0.10 0.00 9.78 v core/U87/A (sky130_fd_sc_hd__clkinv_1)

1 0.00 0.04 0.09 9.87 ^ core/U87/Y (sky130_fd_sc_hd__clkinv_1)

0.04 0.00 9.87 ^ core/U613/A2 (sky130_fd_sc_hd__o21ai_1)

1 0.00 0.07 0.07 9.94 v core/U613/Y (sky130_fd_sc_hd__o21ai_1)

0.07 0.00 9.94 v core/U352/CIN (sky130_fd_sc_hd__fa_1)

1 0.00 0.11 0.66 10.61 v core/U352/COUT (sky130_fd_sc_hd__fa_1)

0.11 0.00 10.61 v core/U348/CIN (sky130_fd_sc_hd__fa_1)

1 0.00 0.11 0.68 11.28 v core/U348/COUT (sky130_fd_sc_hd__fa_1)

0.11 0.00 11.28 v core/U35/CIN (sky130_fd_sc_hd__fa_1)

1 0.00 0.11 0.68 11.96 v core/U35/COUT (sky130_fd_sc_hd__fa_1)

0.11 0.00 11.96 v core/U33/CIN (sky130_fd_sc_hd__fa_1)

2 0.01 0.12 0.70 12.66 v core/U33/COUT (sky130_fd_sc_hd__fa_1)

0.12 0.00 12.66 v core/U81/A1 (sky130_fd_sc_hd__a21o_1)

1 0.00 0.05 0.29 12.95 v core/U81/X (sky130_fd_sc_hd__a21o_1)

0.05 0.00 12.95 v core/U347/CIN (sky130_fd_sc_hd__fa_1)

1 0.00 0.11 0.65 13.61 v core/U347/COUT (sky130_fd_sc_hd__fa_1)

0.11 0.00 13.61 v core/U351/CIN (sky130_fd_sc_hd__fa_1)

2 0.00 0.12 0.82 14.43 v core/U351/SUM (sky130_fd_sc_hd__fa_1)

0.12 0.00 14.43 v core/U3/A (sky130_fd_sc_hd__nand2_1)

15 0.04 0.52 0.45 14.88 ^ core/U3/Y (sky130_fd_sc_hd__nand2_1)

0.52 0.00 14.88 ^ core/U684/B2 (sky130_fd_sc_hd__o22ai_1)

1 0.00 0.17 0.19 15.07 v core/U684/Y (sky130_fd_sc_hd__o22ai_1)

0.17 0.00 15.07 v core/CPU_Xreg_value_a4_reg[27][30]/D (sky130_fd_sc_hd__dfxtp_1)

15.07 data arrival time

0.00 11.00 11.00 clock clk (rise edge)
0.00 11.00 clock network delay (ideal)
0.00 11.00 clock reconvergence pessimism
11.00 ^ core/CPU_Xreg_value_a4_reg[27][30]/CLK (sky130_fd_sc_hd__dfxtp_1)
-0.31 10.69 library setup time
10.69 data required time

10.69 data required time

-15.07 data arrival time

-4.38 slack (VIOLATED)

----- WNS -----

wns max -4.38

----- TNS -----

tns max -367.93

=====
END OF LIBRARY: sky130_fd_sc_hd__ss_n40C_1v76.lib
=====

=====
LIBRARY: sky130_fd_sc_hd__ss_100C_1v40.lib

=====

Warning 1140:
/data/Desktop/work/tools/openlane_working_dir/OpenSTA/vsdbabysoc/lib/sky130_fd_sc_hd__ss_100C_1v40.lib line 1, library sky130_fd_sc_hd__ss_100C_1v40 already exists.

Warning 1140:
/data/Desktop/work/tools/openlane_working_dir/OpenSTA/vsdbabysoc/lib/avsddac.lib line 1, library avsddac already exists.

Warning 1140:
/data/Desktop/work/tools/openlane_working_dir/OpenSTA/vsdbabysoc/lib/avsddac.lib line 1, library avsddac already exists.

Warning 1140:
/data/Desktop/work/tools/openlane_working_dir/OpenSTA/vsdbabysoc/lib/avsdpll.lib line 1, library avsdpll already exists.

Warning 1140:
/data/Desktop/work/tools/openlane_working_dir/OpenSTA/vsdbabysoc/lib/avsdpll.lib line 1, library avsdpll already exists.

----- REPORT CHECKS -----

Warning 168: run_final.tcl line 1, unknown field nets.

Startpoint: core/CPU_imm_a2_reg[10]

(rising edge-triggered flip-flop clocked by clk)

Endpoint: core/CPU_imm_a3_reg[10]

(rising edge-triggered flip-flop clocked by clk)

Path Group: clk

Path Type: min

Fanout	Cap	Slew	Delay	Time	Description
--------	-----	------	-------	------	-------------

0.00	0.00	0.00	clock clk (rise edge)
------	------	------	-----------------------

0.00	0.00	clock network delay (ideal)
------	------	-----------------------------

0.00 0.00 0.00 ^ core/CPU_imm_a2_reg[10]/CLK (sky130_fd_sc_hd__dfxtp_1)
1 0.00 0.05 0.75 0.75 v core/CPU_imm_a2_reg[10]/Q (sky130_fd_sc_hd__dfxtp_1)
0.05 0.00 0.75 v core/CPU_imm_a3_reg[10]/D (sky130_fd_sc_hd__dfxtp_1)
0.75 data arrival time

0.00 0.00 0.00 clock clk (rise edge)
0.00 0.00 clock network delay (ideal)
0.00 0.00 clock reconvergence pessimism

0.00 ^ core/CPU_imm_a3_reg[10]/CLK (sky130_fd_sc_hd__dfxtp_1)
-0.15 -0.15 library hold time
-0.15 data required time

-0.15 data required time
-0.75 data arrival time

0.91 slack (MET)

Startpoint: core/CPU_is_addi_a3_reg
(rising edge-triggered flip-flop clocked by clk)

Endpoint: core/CPU_Xreg_value_a4_reg[27][30]
(rising edge-triggered flip-flop clocked by clk)

Path Group: clk

Path Type: max

Fanout	Cap	Slew	Delay	Time	Description
--------	-----	------	-------	------	-------------

```

0.00 0.00 0.00 clock clk (rise edge)

    0.00 0.00 clock network delay (ideal)

    0.00 0.00 0.00 ^ core/CPU_is_addi_a3_reg/CLK (sky130_fd_sc_hd__dfxtp_1)
2 0.01 0.14 0.91 0.91 ^ core/CPU_is_addi_a3_reg/Q (sky130_fd_sc_hd__dfxtp_1)

    0.14 0.00 0.91 ^ core/U54/A (sky130_fd_sc_hd__clkinv_1)
2 0.01 0.15 0.21 1.13 v core/U54/Y (sky130_fd_sc_hd__clkinv_1)

    0.15 0.00 1.13 v core/U477/A (sky130_fd_sc_hd__nand2_1)
1 0.00 0.13 0.18 1.30 ^ core/U477/Y (sky130_fd_sc_hd__nand2_1)

    0.13 0.00 1.30 ^ core/U51/B (sky130_fd_sc_hd__nand2_2)
32 0.07 0.80 0.71 2.01 v core/U51/Y (sky130_fd_sc_hd__nand2_2)

    0.80 0.00 2.01 v core/U478/B1 (sky130_fd_sc_hd__a22o_1)
1 0.00 0.12 0.80 2.81 v core/U478/X (sky130_fd_sc_hd__a22o_1)

    0.12 0.00 2.81 v core/U479/A (sky130_fd_sc_hd__xor2_1)
2 0.01 0.36 0.47 3.28 v core/U479/X (sky130_fd_sc_hd__xor2_1)

    0.36 0.00 3.28 v core/U569/A2 (sky130_fd_sc_hd__a21oi_1)
2 0.01 0.41 0.58 3.86 ^ core/U569/Y (sky130_fd_sc_hd__a21oi_1)

    0.41 0.00 3.86 ^ core/U571/A2 (sky130_fd_sc_hd__o21ai_1)
2 0.01 0.21 0.36 4.22 v core/U571/Y (sky130_fd_sc_hd__o21ai_1)

    0.21 0.00 4.22 v core/U574/A1 (sky130_fd_sc_hd__a21oi_1)
2 0.01 0.41 0.48 4.70 ^ core/U574/Y (sky130_fd_sc_hd__a21oi_1)

    0.41 0.00 4.70 ^ core/U576/A2 (sky130_fd_sc_hd__o21ai_1)
2 0.01 0.21 0.36 5.05 v core/U576/Y (sky130_fd_sc_hd__o21ai_1)

    0.21 0.00 5.05 v core/U581/A1 (sky130_fd_sc_hd__a21oi_1)
2 0.01 0.41 0.48 5.53 ^ core/U581/Y (sky130_fd_sc_hd__a21oi_1)

    0.41 0.00 5.53 ^ core/U583/A2 (sky130_fd_sc_hd__o21ai_1)
2 0.01 0.21 0.36 5.89 v core/U583/Y (sky130_fd_sc_hd__o21ai_1)

    0.21 0.00 5.89 v core/U587/A1 (sky130_fd_sc_hd__a21oi_1)

```

2 0.01 0.41 0.48 6.37 ^ core/U587/Y (sky130_fd_sc_hd__a21oi_1)
 0.41 0.00 6.37 ^ core/U589/A2 (sky130_fd_sc_hd__o21ai_1)
 2 0.01 0.21 0.36 6.72 v core/U589/Y (sky130_fd_sc_hd__o21ai_1)
 0.21 0.00 6.72 v core/U344/A1 (sky130_fd_sc_hd__a21oi_1)
 2 0.01 0.41 0.48 7.20 ^ core/U344/Y (sky130_fd_sc_hd__a21oi_1)
 0.41 0.00 7.20 ^ core/U213/A2 (sky130_fd_sc_hd__o21ai_1)
 2 0.01 0.21 0.36 7.56 v core/U213/Y (sky130_fd_sc_hd__o21ai_1)
 0.21 0.00 7.56 v core/U343/A1 (sky130_fd_sc_hd__a21oi_1)
 2 0.01 0.41 0.48 8.04 ^ core/U343/Y (sky130_fd_sc_hd__a21oi_1)
 0.41 0.00 8.04 ^ core/U210/A2 (sky130_fd_sc_hd__o21ai_1)
 2 0.01 0.21 0.36 8.39 v core/U210/Y (sky130_fd_sc_hd__o21ai_1)
 0.21 0.00 8.39 v core/U342/A1 (sky130_fd_sc_hd__a21oi_1)
 2 0.01 0.41 0.48 8.87 ^ core/U342/Y (sky130_fd_sc_hd__a21oi_1)
 0.41 0.00 8.87 ^ core/U209/A2 (sky130_fd_sc_hd__o21ai_1)
 2 0.01 0.21 0.36 9.23 v core/U209/Y (sky130_fd_sc_hd__o21ai_1)
 0.21 0.00 9.23 v core/U341/A1 (sky130_fd_sc_hd__a21oi_1)
 2 0.01 0.41 0.48 9.71 ^ core/U341/Y (sky130_fd_sc_hd__a21oi_1)
 0.41 0.00 9.71 ^ core/U215/A2 (sky130_fd_sc_hd__o21ai_1)
 2 0.01 0.21 0.36 10.06 v core/U215/Y (sky130_fd_sc_hd__o21ai_1)
 0.21 0.00 10.06 v core/U96/A1 (sky130_fd_sc_hd__a21oi_1)
 1 0.00 0.10 0.49 10.55 v core/U96/X (sky130_fd_sc_hd__a21oi_1)
 0.10 0.00 10.55 v core/U942/CIN (sky130_fd_sc_hd__fa_1)
 2 0.01 0.25 1.32 11.87 v core/U942/COUT (sky130_fd_sc_hd__fa_1)
 0.25 0.00 11.87 v core/U94/A1 (sky130_fd_sc_hd__a21oi_1)
 1 0.00 0.10 0.51 12.38 v core/U94/X (sky130_fd_sc_hd__a21oi_1)
 0.10 0.00 12.38 v core/U905/CIN (sky130_fd_sc_hd__fa_1)
 1 0.00 0.22 1.27 13.65 v core/U905/COUT (sky130_fd_sc_hd__fa_1)

0.22 0.00 13.65 v core/U887/CIN (sky130_fd_sc_hd__fa_1)

1 0.00 0.22 1.33 14.98 v core/U887/COUT (sky130_fd_sc_hd__fa_1)

0.22 0.00 14.98 v core/U1370/CIN (sky130_fd_sc_hd__fa_1)

1 0.00 0.23 1.34 16.32 v core/U1370/COUT (sky130_fd_sc_hd__fa_1)

0.23 0.00 16.32 v core/U36/CIN (sky130_fd_sc_hd__fa_2)

2 0.01 0.19 1.13 17.45 v core/U36/COUT (sky130_fd_sc_hd__fa_2)

0.19 0.00 17.45 v core/U87/A (sky130_fd_sc_hd__clkinv_1)

1 0.00 0.07 0.13 17.58 ^ core/U87/Y (sky130_fd_sc_hd__clkinv_1)

0.07 0.00 17.58 ^ core/U613/A2 (sky130_fd_sc_hd__o21ai_1)

1 0.00 0.16 0.17 17.75 v core/U613/Y (sky130_fd_sc_hd__o21ai_1)

0.16 0.00 17.75 v core/U352/CIN (sky130_fd_sc_hd__fa_1)

1 0.00 0.22 1.30 19.05 v core/U352/COUT (sky130_fd_sc_hd__fa_1)

0.22 0.00 19.05 v core/U348/CIN (sky130_fd_sc_hd__fa_1)

1 0.00 0.22 1.33 20.38 v core/U348/COUT (sky130_fd_sc_hd__fa_1)

0.22 0.00 20.38 v core/U35/CIN (sky130_fd_sc_hd__fa_1)

1 0.00 0.22 1.33 21.70 v core/U35/COUT (sky130_fd_sc_hd__fa_1)

0.22 0.00 21.70 v core/U33/CIN (sky130_fd_sc_hd__fa_1)

2 0.01 0.25 1.37 23.08 v core/U33/COUT (sky130_fd_sc_hd__fa_1)

0.25 0.00 23.08 v core/U81/A1 (sky130_fd_sc_hd__a21o_1)

1 0.00 0.10 0.51 23.59 v core/U81/X (sky130_fd_sc_hd__a21o_1)

0.10 0.00 23.59 v core/U347/CIN (sky130_fd_sc_hd__fa_1)

1 0.00 0.22 1.27 24.86 v core/U347/COUT (sky130_fd_sc_hd__fa_1)

0.22 0.00 24.86 v core/U351/CIN (sky130_fd_sc_hd__fa_1)

2 0.00 0.25 1.63 26.49 v core/U351/SUM (sky130_fd_sc_hd__fa_1)

0.25 0.00 26.49 v core/U3/A (sky130_fd_sc_hd__nand2_1)

15 0.04 0.74 0.66 27.15 ^ core/U3/Y (sky130_fd_sc_hd__nand2_1)

0.74 0.00 27.15 ^ core/U684/B2 (sky130_fd_sc_hd__o22ai_1)

1 0.00 0.30 0.39 27.54 v core/U684/Y (sky130_fd_sc_hd__o22ai_1)
0.30 0.00 27.54 v core/CPU_Xreg_value_a4_reg[27][30]/D (sky130_fd_sc_hd__dfxtp_1)
27.54 data arrival time

0.00 11.00 11.00 clock clk (rise edge)
0.00 11.00 clock network delay (ideal)
0.00 11.00 clock reconvergence pessimism
11.00 ^ core/CPU_Xreg_value_a4_reg[27][30]/CLK (sky130_fd_sc_hd__dfxtp_1)
-0.58 10.42 library setup time
10.42 data required time

10.42 data required time
-27.54 data arrival time

-17.12 slack (VIOLATED)

----- WNS -----

wns max -17.12

----- TNS -----

tns max -2599.51

=====

END OF LIBRARY: sky130_fd_sc_hd__ss_100C_1v40.lib

=====

=====

LIBRARY: sky130_fd_sc_hd__ss_100C_1v60.lib

=====

Warning 1140:
/data/Desktop/work/tools/openlane_working_dir/OpenSTA/vsdbabysoc/lib/sky130_fd_sc_hd__ss_100C_1v60.lib line 1, library sky130_fd_sc_hd__ss_100C_1v60 already exists.

Warning 1140:
/data/Desktop/work/tools/openlane_working_dir/OpenSTA/vsdbabysoc/lib/avsddac.lib line 1, library avsddac already exists.

Warning 1140:
/data/Desktop/work/tools/openlane_working_dir/OpenSTA/vsdbabysoc/lib/avsddac.lib line 1, library avsddac already exists.

Warning 1140:
/data/Desktop/work/tools/openlane_working_dir/OpenSTA/vsdbabysoc/lib/avsdpll.lib line 1, library avsdpll already exists.

Warning 1140:
/data/Desktop/work/tools/openlane_working_dir/OpenSTA/vsdbabysoc/lib/avsdpll.lib line 1, library avsdpll already exists.

----- REPORT CHECKS -----

Warning 168: run_final.tcl line 1, unknown field nets.

Startpoint: core/CPU_imm_a2_reg[10]

(rising edge-triggered flip-flop clocked by clk)

Endpoint: core/CPU_imm_a3_reg[10]

(rising edge-triggered flip-flop clocked by clk)

Path Group: clk

Path Type: min

Fanout	Cap	Slew	Delay	Time	Description
--------	-----	------	-------	------	-------------

	0.00	0.00	0.00		clock clk (rise edge)
		0.00	0.00		clock network delay (ideal)
	0.00	0.00	0.00		^ core/CPU_imm_a2_reg[10]/CLK (sky130_fd_sc_hd__dfxtp_1)
1	0.00	0.04	0.53	0.53	v core/CPU_imm_a2_reg[10]/Q (sky130_fd_sc_hd__dfxtp_1)
	0.04	0.00	0.53		v core/CPU_imm_a3_reg[10]/D (sky130_fd_sc_hd__dfxtp_1)
			0.53		data arrival time

	0.00	0.00	0.00		clock clk (rise edge)
		0.00	0.00		clock network delay (ideal)
		0.00	0.00		clock reconvergence pessimism
			0.00		^ core/CPU_imm_a3_reg[10]/CLK (sky130_fd_sc_hd__dfxtp_1)
	-0.11	-0.11			library hold time
			-0.11		data required time

			-0.11		data required time
			-0.53		data arrival time

				0.64	slack (MET)
--	--	--	--	------	-------------

Startpoint: core/CPU_is_addi_a3_reg

(rising edge-triggered flip-flop clocked by clk)

Endpoint: core/CPU_Xreg_value_a4_reg[27][30]

(rising edge-triggered flip-flop clocked by clk)

Path Group: clk

Path Type: max

Fanout	Cap	Slew	Delay	Time	Description
--------	-----	------	-------	------	-------------

	0.00	0.00	0.00		clock clk (rise edge)
	0.00	0.00			clock network delay (ideal)
	0.00	0.00	0.00		^ core/CPU_is_addi_a3_reg/CLK (sky130_fd_sc_hd__dfxtp_1)
2	0.01	0.11	0.62	0.62	^ core/CPU_is_addi_a3_reg/Q (sky130_fd_sc_hd__dfxtp_1)
	0.11	0.00	0.62		^ core/U54/A (sky130_fd_sc_hd__clkinv_1)
2	0.01	0.12	0.15	0.78	v core/U54/Y (sky130_fd_sc_hd__clkinv_1)
	0.12	0.00	0.78		v core/U477/A (sky130_fd_sc_hd__nand2_1)
1	0.00	0.10	0.13	0.91	^ core/U477/Y (sky130_fd_sc_hd__nand2_1)
	0.10	0.00	0.91		^ core/U51/B (sky130_fd_sc_hd__nand2_2)
32	0.07	0.60	0.51	1.42	v core/U51/Y (sky130_fd_sc_hd__nand2_2)
	0.60	0.00	1.42		v core/U478/B1 (sky130_fd_sc_hd__a22o_1)
1	0.00	0.09	0.56	1.98	v core/U478/X (sky130_fd_sc_hd__a22o_1)
	0.09	0.00	1.98		v core/U479/A (sky130_fd_sc_hd__xor2_1)
2	0.01	0.29	0.33	2.31	v core/U479/X (sky130_fd_sc_hd__xor2_1)
	0.29	0.00	2.31		v core/U569/A2 (sky130_fd_sc_hd__a21oi_1)
2	0.01	0.31	0.42	2.72	^ core/U569/Y (sky130_fd_sc_hd__a21oi_1)
	0.31	0.00	2.72		^ core/U571/A2 (sky130_fd_sc_hd__o21ai_1)
2	0.01	0.16	0.24	2.97	v core/U571/Y (sky130_fd_sc_hd__o21ai_1)
	0.16	0.00	2.97		v core/U574/A1 (sky130_fd_sc_hd__a21oi_1)
2	0.01	0.31	0.34	3.31	^ core/U574/Y (sky130_fd_sc_hd__a21oi_1)
	0.31	0.00	3.31		^ core/U576/A2 (sky130_fd_sc_hd__o21ai_1)
2	0.01	0.16	0.24	3.55	v core/U576/Y (sky130_fd_sc_hd__o21ai_1)
	0.16	0.00	3.55		v core/U581/A1 (sky130_fd_sc_hd__a21oi_1)

2 0.01 0.31 0.34 3.89 ^ core/U581/Y (sky130_fd_sc_hd__a21oi_1)
 0.31 0.00 3.89 ^ core/U583/A2 (sky130_fd_sc_hd__o21ai_1)
 2 0.01 0.16 0.24 4.13 v core/U583/Y (sky130_fd_sc_hd__o21ai_1)
 0.16 0.00 4.13 v core/U587/A1 (sky130_fd_sc_hd__a21oi_1)
 2 0.01 0.31 0.34 4.47 ^ core/U587/Y (sky130_fd_sc_hd__a21oi_1)
 0.31 0.00 4.47 ^ core/U589/A2 (sky130_fd_sc_hd__o21ai_1)
 2 0.01 0.16 0.24 4.72 v core/U589/Y (sky130_fd_sc_hd__o21ai_1)
 0.16 0.00 4.72 v core/U344/A1 (sky130_fd_sc_hd__a21oi_1)
 2 0.01 0.31 0.34 5.06 ^ core/U344/Y (sky130_fd_sc_hd__a21oi_1)
 0.31 0.00 5.06 ^ core/U213/A2 (sky130_fd_sc_hd__o21ai_1)
 2 0.01 0.16 0.24 5.30 v core/U213/Y (sky130_fd_sc_hd__o21ai_1)
 0.16 0.00 5.30 v core/U343/A1 (sky130_fd_sc_hd__a21oi_1)
 2 0.01 0.31 0.34 5.64 ^ core/U343/Y (sky130_fd_sc_hd__a21oi_1)
 0.31 0.00 5.64 ^ core/U210/A2 (sky130_fd_sc_hd__o21ai_1)
 2 0.01 0.16 0.24 5.88 v core/U210/Y (sky130_fd_sc_hd__o21ai_1)
 0.16 0.00 5.88 v core/U342/A1 (sky130_fd_sc_hd__a21oi_1)
 2 0.01 0.31 0.34 6.22 ^ core/U342/Y (sky130_fd_sc_hd__a21oi_1)
 0.31 0.00 6.22 ^ core/U209/A2 (sky130_fd_sc_hd__o21ai_1)
 2 0.01 0.16 0.24 6.46 v core/U209/Y (sky130_fd_sc_hd__o21ai_1)
 0.16 0.00 6.46 v core/U341/A1 (sky130_fd_sc_hd__a21oi_1)
 2 0.01 0.31 0.34 6.81 ^ core/U341/Y (sky130_fd_sc_hd__a21oi_1)
 0.31 0.00 6.81 ^ core/U215/A2 (sky130_fd_sc_hd__o21ai_1)
 2 0.01 0.16 0.24 7.05 v core/U215/Y (sky130_fd_sc_hd__o21ai_1)
 0.16 0.00 7.05 v core/U96/A1 (sky130_fd_sc_hd__a21o_1)
 1 0.00 0.08 0.34 7.38 v core/U96/X (sky130_fd_sc_hd__a21o_1)
 0.08 0.00 7.38 v core/U942/CIN (sky130_fd_sc_hd__fa_1)
 2 0.01 0.18 0.86 8.24 v core/U942/COUT (sky130_fd_sc_hd__fa_1)

0.18 0.00 8.24 v core/U94/A1 (sky130_fd_sc_hd__a21o_1)

1 0.00 0.08 0.35 8.59 v core/U94/X (sky130_fd_sc_hd__a21o_1)

0.08 0.00 8.59 v core/U905/CIN (sky130_fd_sc_hd__fa_1)

1 0.00 0.16 0.83 9.42 v core/U905/COUT (sky130_fd_sc_hd__fa_1)

0.16 0.00 9.42 v core/U887/CIN (sky130_fd_sc_hd__fa_1)

1 0.00 0.16 0.86 10.28 v core/U887/COUT (sky130_fd_sc_hd__fa_1)

0.16 0.00 10.28 v core/U1370/CIN (sky130_fd_sc_hd__fa_1)

1 0.00 0.16 0.87 11.14 v core/U1370/COUT (sky130_fd_sc_hd__fa_1)

0.16 0.00 11.14 v core/U36/CIN (sky130_fd_sc_hd__fa_2)

2 0.01 0.14 0.74 11.89 v core/U36/COUT (sky130_fd_sc_hd__fa_2)

0.14 0.00 11.89 v core/U87/A (sky130_fd_sc_hd__clkinv_1)

1 0.00 0.05 0.09 11.98 ^ core/U87/Y (sky130_fd_sc_hd__clkinv_1)

0.05 0.00 11.98 ^ core/U613/A2 (sky130_fd_sc_hd__o21ai_1)

1 0.00 0.12 0.12 12.10 v core/U613/Y (sky130_fd_sc_hd__o21ai_1)

0.12 0.00 12.10 v core/U352/CIN (sky130_fd_sc_hd__fa_1)

1 0.00 0.16 0.84 12.94 v core/U352/COUT (sky130_fd_sc_hd__fa_1)

0.16 0.00 12.94 v core/U348/CIN (sky130_fd_sc_hd__fa_1)

1 0.00 0.16 0.86 13.80 v core/U348/COUT (sky130_fd_sc_hd__fa_1)

0.16 0.00 13.80 v core/U35/CIN (sky130_fd_sc_hd__fa_1)

1 0.00 0.16 0.86 14.66 v core/U35/COUT (sky130_fd_sc_hd__fa_1)

0.16 0.00 14.66 v core/U33/CIN (sky130_fd_sc_hd__fa_1)

2 0.01 0.18 0.89 15.55 v core/U33/COUT (sky130_fd_sc_hd__fa_1)

0.18 0.00 15.55 v core/U81/A1 (sky130_fd_sc_hd__a21o_1)

1 0.00 0.08 0.35 15.90 v core/U81/X (sky130_fd_sc_hd__a21o_1)

0.08 0.00 15.90 v core/U347/CIN (sky130_fd_sc_hd__fa_1)

1 0.00 0.16 0.83 16.73 v core/U347/COUT (sky130_fd_sc_hd__fa_1)

0.16 0.00 16.73 v core/U351/CIN (sky130_fd_sc_hd__fa_1)

```

2  0.01  0.18  1.04  17.77 v core/U351/SUM (sky130_fd_sc_hd__fa_1)
    0.18  0.00  17.77 v core/U3/A (sky130_fd_sc_hd__nand2_1)
15  0.04  0.55  0.48  18.25 ^ core/U3/Y (sky130_fd_sc_hd__nand2_1)
    0.55  0.00  18.25 ^ core/U684/B2 (sky130_fd_sc_hd__o22ai_1)
1  0.00  0.24  0.26  18.51 v core/U684/Y (sky130_fd_sc_hd__o22ai_1)
    0.24  0.00  18.51 v core/CPU_Xreg_value_a4_reg[27][30]/D (sky130_fd_sc_hd__dfxtp_1)
    18.51  data arrival time

```

```

0.00  11.00  11.00  clock clk (rise edge)
    0.00  11.00  clock network delay (ideal)
    0.00  11.00  clock reconvergence pessimism
    11.00 ^ core/CPU_Xreg_value_a4_reg[27][30]/CLK (sky130_fd_sc_hd__dfxtp_1)
-0.36  10.64  library setup time
    10.64  data required time

```

```

    10.64  data required time
    -18.51  data arrival time

```

```

    -7.87  slack (VIOLATED)

```

----- WNS -----

wns max -7.87

----- TNS -----

tns max -892.70

=====

END OF LIBRARY: sky130_fd_sc_hd__ss_100C_1v60.lib

=====

=====

LIBRARY: sky130_fd_sc_hd__ff_n40C_1v56.lib

=====

Warning 1140:
/data/Desktop/work/tools/openlane_working_dir/OpenSTA/vsdbabysoc/lib/sky130_fd_sc_hd__ff_n40C_1v56.lib line 1, library sky130_fd_sc_hd__ff_n40C_1v56 already exists.

Warning 1140:
/data/Desktop/work/tools/openlane_working_dir/OpenSTA/vsdbabysoc/lib/avsddac.lib line 1, library avsddac already exists.

Warning 1140:
/data/Desktop/work/tools/openlane_working_dir/OpenSTA/vsdbabysoc/lib/avsddac.lib line 1, library avsddac already exists.

Warning 1140:
/data/Desktop/work/tools/openlane_working_dir/OpenSTA/vsdbabysoc/lib/avsdpll.lib line 1, library avsdpll already exists.

Warning 1140:
/data/Desktop/work/tools/openlane_working_dir/OpenSTA/vsdbabysoc/lib/avsdpll.lib line 1, library avsdpll already exists.

----- REPORT CHECKS -----

Warning 168: run_final.tcl line 1, unknown field nets.

Startpoint: core/CPU_imm_a2_reg[10]

(rising edge-triggered flip-flop clocked by clk)

Endpoint: core/CPU_imm_a3_reg[10]

(rising edge-triggered flip-flop clocked by clk)

Path Group: clk

Path Type: min

Fanout	Cap	Slew	Delay	Time	Description
--------	-----	------	-------	------	-------------

	0.00	0.00	0.00		clock clk (rise edge)
		0.00	0.00		clock network delay (ideal)
	0.00	0.00	0.00		^ core/CPU_imm_a2_reg[10]/CLK (sky130_fd_sc_hd__dfxtp_1)
1	0.00	0.03	0.26	0.26	^ core/CPU_imm_a2_reg[10]/Q (sky130_fd_sc_hd__dfxtp_1)
	0.03	0.00	0.26		^ core/CPU_imm_a3_reg[10]/D (sky130_fd_sc_hd__dfxtp_1)
			0.26		data arrival time
	0.00	0.00	0.00		clock clk (rise edge)
		0.00	0.00		clock network delay (ideal)
	0.00	0.00			clock reconvergence pessimism
			0.00		^ core/CPU_imm_a3_reg[10]/CLK (sky130_fd_sc_hd__dfxtp_1)
	-0.03	-0.03			library hold time
		-0.03			data required time

			-0.03		data required time
			-0.26		data arrival time

			0.29		slack (MET)

Startpoint: core/CPU_is_addi_a3_reg

(rising edge-triggered flip-flop clocked by clk)

Endpoint: core/CPU_Xreg_value_a4_reg[16][31]

(rising edge-triggered flip-flop clocked by clk)

Path Group: clk

Path Type: max

Fanout	Cap	Slew	Delay	Time	Description
--------	-----	------	-------	------	-------------

	0.00	0.00	0.00		clock clk (rise edge)
		0.00	0.00		clock network delay (ideal)
	0.00	0.00	0.00		^ core/CPU_is_addi_a3_reg/CLK (sky130_fd_sc_hd__dfxtp_1)
2	0.00	0.03	0.27	0.27	v core/CPU_is_addi_a3_reg/Q (sky130_fd_sc_hd__dfxtp_1)
	0.03	0.00	0.27		v core/U474/A (sky130_fd_sc_hd__nor2_1)
2	0.00	0.12	0.13	0.39	^ core/U474/Y (sky130_fd_sc_hd__nor2_1)
	0.12	0.00	0.39		^ core/U476/A (sky130_fd_sc_hd__nand2_1)
2	0.01	0.06	0.08	0.48	v core/U476/Y (sky130_fd_sc_hd__nand2_1)
	0.06	0.00	0.48		v core/U9/A (sky130_fd_sc_hd__inv_2)
34	0.14	0.71	0.54	1.01	^ core/U9/Y (sky130_fd_sc_hd__inv_2)
	0.71	0.00	1.01		^ core/U479/B (sky130_fd_sc_hd__xor2_1)
2	0.01	0.15	0.15	1.16	v core/U479/X (sky130_fd_sc_hd__xor2_1)
	0.15	0.00	1.16		v core/U569/A2 (sky130_fd_sc_hd__a21oi_1)
2	0.01	0.20	0.27	1.43	^ core/U569/Y (sky130_fd_sc_hd__a21oi_1)
	0.20	0.00	1.43		^ core/U571/A2 (sky130_fd_sc_hd__o21ai_1)
2	0.01	0.06	0.10	1.53	v core/U571/Y (sky130_fd_sc_hd__o21ai_1)
	0.06	0.00	1.53		v core/U574/A1 (sky130_fd_sc_hd__a21oi_1)
2	0.01	0.20	0.21	1.74	^ core/U574/Y (sky130_fd_sc_hd__a21oi_1)
	0.20	0.00	1.74		^ core/U576/A2 (sky130_fd_sc_hd__o21ai_1)

2 0.01 0.06 0.10 1.84 v core/U576/Y (sky130_fd_sc_hd__o21ai_1)
 0.06 0.00 1.84 v core/U581/A1 (sky130_fd_sc_hd__a21oi_1)
 2 0.01 0.20 0.21 2.04 ^ core/U581/Y (sky130_fd_sc_hd__a21oi_1)
 0.20 0.00 2.04 ^ core/U583/A2 (sky130_fd_sc_hd__o21ai_1)
 2 0.01 0.06 0.10 2.14 v core/U583/Y (sky130_fd_sc_hd__o21ai_1)
 0.06 0.00 2.14 v core/U587/A1 (sky130_fd_sc_hd__a21oi_1)
 2 0.01 0.20 0.21 2.35 ^ core/U587/Y (sky130_fd_sc_hd__a21oi_1)
 0.20 0.00 2.35 ^ core/U589/A2 (sky130_fd_sc_hd__o21ai_1)
 2 0.01 0.06 0.10 2.45 v core/U589/Y (sky130_fd_sc_hd__o21ai_1)
 0.06 0.00 2.45 v core/U344/A1 (sky130_fd_sc_hd__a21oi_1)
 2 0.01 0.20 0.21 2.65 ^ core/U344/Y (sky130_fd_sc_hd__a21oi_1)
 0.20 0.00 2.65 ^ core/U213/A2 (sky130_fd_sc_hd__o21ai_1)
 2 0.01 0.06 0.10 2.75 v core/U213/Y (sky130_fd_sc_hd__o21ai_1)
 0.06 0.00 2.75 v core/U343/A1 (sky130_fd_sc_hd__a21oi_1)
 2 0.01 0.20 0.21 2.96 ^ core/U343/Y (sky130_fd_sc_hd__a21oi_1)
 0.20 0.00 2.96 ^ core/U210/A2 (sky130_fd_sc_hd__o21ai_1)
 2 0.01 0.06 0.10 3.06 v core/U210/Y (sky130_fd_sc_hd__o21ai_1)
 0.06 0.00 3.06 v core/U342/A1 (sky130_fd_sc_hd__a21oi_1)
 2 0.01 0.20 0.21 3.26 ^ core/U342/Y (sky130_fd_sc_hd__a21oi_1)
 0.20 0.00 3.26 ^ core/U209/A2 (sky130_fd_sc_hd__o21ai_1)
 2 0.01 0.06 0.10 3.36 v core/U209/Y (sky130_fd_sc_hd__o21ai_1)
 0.06 0.00 3.36 v core/U341/A1 (sky130_fd_sc_hd__a21oi_1)
 2 0.01 0.20 0.21 3.57 ^ core/U341/Y (sky130_fd_sc_hd__a21oi_1)
 0.20 0.00 3.57 ^ core/U215/A2 (sky130_fd_sc_hd__o21ai_1)
 2 0.01 0.06 0.10 3.67 v core/U215/Y (sky130_fd_sc_hd__o21ai_1)
 0.06 0.00 3.67 v core/U96/A1 (sky130_fd_sc_hd__a21o_1)
 1 0.00 0.04 0.17 3.84 v core/U96/X (sky130_fd_sc_hd__a21o_1)

0.04 0.00 3.84 v core/U942/CIN (sky130_fd_sc_hd__fa_1)

2 0.01 0.09 0.35 4.19 v core/U942/COUT (sky130_fd_sc_hd__fa_1)

0.09 0.00 4.19 v core/U94/A1 (sky130_fd_sc_hd__a21o_1)

1 0.00 0.04 0.18 4.38 v core/U94/X (sky130_fd_sc_hd__a21o_1)

0.04 0.00 4.38 v core/U905/CIN (sky130_fd_sc_hd__fa_1)

1 0.00 0.08 0.33 4.71 v core/U905/COUT (sky130_fd_sc_hd__fa_1)

0.08 0.00 4.71 v core/U887/CIN (sky130_fd_sc_hd__fa_1)

1 0.00 0.08 0.35 5.06 v core/U887/COUT (sky130_fd_sc_hd__fa_1)

0.08 0.00 5.06 v core/U1370/CIN (sky130_fd_sc_hd__fa_1)

1 0.00 0.08 0.35 5.41 v core/U1370/COUT (sky130_fd_sc_hd__fa_1)

0.08 0.00 5.41 v core/U36/CIN (sky130_fd_sc_hd__fa_2)

2 0.01 0.07 0.34 5.74 v core/U36/COUT (sky130_fd_sc_hd__fa_2)

0.07 0.00 5.74 v core/U87/A (sky130_fd_sc_hd__clkinv_1)

1 0.00 0.03 0.06 5.80 ^ core/U87/Y (sky130_fd_sc_hd__clkinv_1)

0.03 0.00 5.80 ^ core/U613/A2 (sky130_fd_sc_hd__o21ai_1)

1 0.00 0.05 0.05 5.85 v core/U613/Y (sky130_fd_sc_hd__o21ai_1)

0.05 0.00 5.85 v core/U352/CIN (sky130_fd_sc_hd__fa_1)

1 0.00 0.08 0.34 6.18 v core/U352/COUT (sky130_fd_sc_hd__fa_1)

0.08 0.00 6.18 v core/U348/CIN (sky130_fd_sc_hd__fa_1)

1 0.00 0.08 0.35 6.53 v core/U348/COUT (sky130_fd_sc_hd__fa_1)

0.08 0.00 6.53 v core/U35/CIN (sky130_fd_sc_hd__fa_1)

1 0.00 0.08 0.35 6.88 v core/U35/COUT (sky130_fd_sc_hd__fa_1)

0.08 0.00 6.88 v core/U33/CIN (sky130_fd_sc_hd__fa_1)

2 0.01 0.09 0.36 7.24 v core/U33/COUT (sky130_fd_sc_hd__fa_1)

0.09 0.00 7.24 v core/U81/A1 (sky130_fd_sc_hd__a21o_1)

1 0.00 0.04 0.18 7.42 v core/U81/X (sky130_fd_sc_hd__a21o_1)

0.04 0.00 7.42 v core/U347/CIN (sky130_fd_sc_hd__fa_1)

```

1  0.00  0.08  0.33  7.76 v core/U347/COUT (sky130_fd_sc_hd__fa_1)
    0.08  0.00  7.76 v core/U351/CIN (sky130_fd_sc_hd__fa_1)
1  0.00  0.07  0.34  8.10 v core/U351/COUT (sky130_fd_sc_hd__fa_1)
    0.07  0.00  8.10 v core/U617/B (sky130_fd_sc_hd__xor2_1)
2  0.01  0.22  0.22  8.32 ^ core/U617/X (sky130_fd_sc_hd__xor2_1)
    0.22  0.00  8.32 ^ core/U618/A (sky130_fd_sc_hd__nand2_2)
15 0.03  0.20  0.16  8.48 v core/U618/Y (sky130_fd_sc_hd__nand2_2)
    0.20  0.00  8.48 v core/U635/B2 (sky130_fd_sc_hd__o22ai_1)
1  0.00  0.15  0.16  8.64 ^ core/U635/Y (sky130_fd_sc_hd__o22ai_1)
    0.15  0.00  8.64 ^ core/CPU_Xreg_value_a4_reg[16][31]/D (sky130_fd_sc_hd__dfxtp_1)
    8.64  data arrival time

```

```

0.00 11.00 11.00 clock clk (rise edge)
    0.00 11.00 clock network delay (ideal)
    0.00 11.00 clock reconvergence pessimism
    11.00 ^ core/CPU_Xreg_value_a4_reg[16][31]/CLK (sky130_fd_sc_hd__dfxtp_1)
-0.08 10.92 library setup time
    10.92 data required time

```

```

-----
10.92 data required time

```

```

-8.64 data arrival time
-----

```

```

2.28 slack (MET)

```

```

----- WNS -----

```

wns max 0.00

----- TNS -----

tns max 0.00

=====

END OF LIBRARY: sky130_fd_sc_hd__ff_n40C_1v56.lib

=====

=====

LIBRARY: sky130_fd_sc_hd__ff_n40C_1v65.lib

=====

Warning 1140:
/data/Desktop/work/tools/openlane_working_dir/OpenSTA/vsdbabysoc/lib/sky130_fd_sc_hd__ff_n40C_1v65.lib line 1, library sky130_fd_sc_hd__ff_n40C_1v65 already exists.

Warning 1140:
/data/Desktop/work/tools/openlane_working_dir/OpenSTA/vsdbabysoc/lib/avsdac.lib line 1, library avsdac already exists.

Warning 1140:
/data/Desktop/work/tools/openlane_working_dir/OpenSTA/vsdbabysoc/lib/avsdac.lib line 1, library avsdac already exists.

Warning 1140:
/data/Desktop/work/tools/openlane_working_dir/OpenSTA/vsdbabysoc/lib/avsdpll.lib line 1, library avsdpll already exists.

Warning 1140:
/data/Desktop/work/tools/openlane_working_dir/OpenSTA/vsdbabysoc/lib/avsdpll.lib line 1, library avsdpll already exists.

----- REPORT CHECKS -----

Warning 168: run_final.tcl line 1, unknown field nets.

Startpoint: core/CPU_imm_a2_reg[10]

(rising edge-triggered flip-flop clocked by clk)

Endpoint: core/CPU_imm_a3_reg[10]

(rising edge-triggered flip-flop clocked by clk)

Path Group: clk

Path Type: min

Fanout	Cap	Slew	Delay	Time	Description
--------	-----	------	-------	------	-------------

	0.00	0.00	0.00		clock clk (rise edge)
		0.00	0.00		clock network delay (ideal)
	0.00	0.00	0.00		^ core/CPU_imm_a2_reg[10]/CLK (sky130_fd_sc_hd__dfxtp_1)
1	0.00	0.03	0.23	0.23	^ core/CPU_imm_a2_reg[10]/Q (sky130_fd_sc_hd__dfxtp_1)
	0.03	0.00	0.23		^ core/CPU_imm_a3_reg[10]/D (sky130_fd_sc_hd__dfxtp_1)
			0.23		data arrival time
	0.00	0.00	0.00		clock clk (rise edge)
		0.00	0.00		clock network delay (ideal)
	0.00	0.00			clock reconvergence pessimism
			0.00		^ core/CPU_imm_a3_reg[10]/CLK (sky130_fd_sc_hd__dfxtp_1)
	-0.02	-0.02			library hold time
		-0.02			data required time

		-0.02			data required time
		-0.23			data arrival time

0.26 slack (MET)

Startpoint: core/CPU_is_addi_a3_reg

(rising edge-triggered flip-flop clocked by clk)

Endpoint: core/CPU_Xreg_value_a4_reg[16][31]

(rising edge-triggered flip-flop clocked by clk)

Path Group: clk

Path Type: max

Fanout	Cap	Slew	Delay	Time	Description
--------	-----	------	-------	------	-------------

	0.00	0.00	0.00		clock clk (rise edge)
	0.00	0.00			clock network delay (ideal)
	0.00	0.00	0.00		^ core/CPU_is_addi_a3_reg/CLK (sky130_fd_sc_hd__dfxtp_1)
2	0.00	0.03	0.24	0.24	v core/CPU_is_addi_a3_reg/Q (sky130_fd_sc_hd__dfxtp_1)
	0.03	0.00	0.24		v core/U474/A (sky130_fd_sc_hd__nor2_1)
2	0.00	0.10	0.11	0.35	^ core/U474/Y (sky130_fd_sc_hd__nor2_1)
	0.10	0.00	0.35		^ core/U476/A (sky130_fd_sc_hd__nand2_1)
2	0.01	0.06	0.07	0.42	v core/U476/Y (sky130_fd_sc_hd__nand2_1)
	0.06	0.00	0.42		v core/U9/A (sky130_fd_sc_hd__inv_2)
34	0.14	0.64	0.48	0.90	^ core/U9/Y (sky130_fd_sc_hd__inv_2)
	0.64	0.00	0.90		^ core/U479/B (sky130_fd_sc_hd__xor2_1)
2	0.01	0.14	0.12	1.02	v core/U479/X (sky130_fd_sc_hd__xor2_1)
	0.14	0.00	1.02		v core/U569/A2 (sky130_fd_sc_hd__a21oi_1)
2	0.01	0.18	0.23	1.26	^ core/U569/Y (sky130_fd_sc_hd__a21oi_1)
	0.18	0.00	1.26		^ core/U571/A2 (sky130_fd_sc_hd__o21ai_1)

2 0.01 0.06 0.09 1.34 v core/U571/Y (sky130_fd_sc_hd__o21ai_1)
 0.06 0.00 1.34 v core/U574/A1 (sky130_fd_sc_hd__a21oi_1)
 2 0.01 0.18 0.18 1.53 ^ core/U574/Y (sky130_fd_sc_hd__a21oi_1)
 0.18 0.00 1.53 ^ core/U576/A2 (sky130_fd_sc_hd__o21ai_1)
 2 0.01 0.06 0.09 1.61 v core/U576/Y (sky130_fd_sc_hd__o21ai_1)
 0.06 0.00 1.61 v core/U581/A1 (sky130_fd_sc_hd__a21oi_1)
 2 0.01 0.18 0.18 1.79 ^ core/U581/Y (sky130_fd_sc_hd__a21oi_1)
 0.18 0.00 1.79 ^ core/U583/A2 (sky130_fd_sc_hd__o21ai_1)
 2 0.01 0.06 0.09 1.88 v core/U583/Y (sky130_fd_sc_hd__o21ai_1)
 0.06 0.00 1.88 v core/U587/A1 (sky130_fd_sc_hd__a21oi_1)
 2 0.01 0.18 0.18 2.06 ^ core/U587/Y (sky130_fd_sc_hd__a21oi_1)
 0.18 0.00 2.06 ^ core/U589/A2 (sky130_fd_sc_hd__o21ai_1)
 2 0.01 0.06 0.09 2.15 v core/U589/Y (sky130_fd_sc_hd__o21ai_1)
 0.06 0.00 2.15 v core/U344/A1 (sky130_fd_sc_hd__a21oi_1)
 2 0.01 0.18 0.18 2.33 ^ core/U344/Y (sky130_fd_sc_hd__a21oi_1)
 0.18 0.00 2.33 ^ core/U213/A2 (sky130_fd_sc_hd__o21ai_1)
 2 0.01 0.06 0.09 2.41 v core/U213/Y (sky130_fd_sc_hd__o21ai_1)
 0.06 0.00 2.41 v core/U343/A1 (sky130_fd_sc_hd__a21oi_1)
 2 0.01 0.18 0.18 2.59 ^ core/U343/Y (sky130_fd_sc_hd__a21oi_1)
 0.18 0.00 2.59 ^ core/U210/A2 (sky130_fd_sc_hd__o21ai_1)
 2 0.01 0.06 0.09 2.68 v core/U210/Y (sky130_fd_sc_hd__o21ai_1)
 0.06 0.00 2.68 v core/U342/A1 (sky130_fd_sc_hd__a21oi_1)
 2 0.01 0.18 0.18 2.86 ^ core/U342/Y (sky130_fd_sc_hd__a21oi_1)
 0.18 0.00 2.86 ^ core/U209/A2 (sky130_fd_sc_hd__o21ai_1)
 2 0.01 0.06 0.09 2.95 v core/U209/Y (sky130_fd_sc_hd__o21ai_1)
 0.06 0.00 2.95 v core/U341/A1 (sky130_fd_sc_hd__a21oi_1)
 2 0.01 0.18 0.18 3.13 ^ core/U341/Y (sky130_fd_sc_hd__a21oi_1)

0.18 0.00 3.13 ^ core/U215/A2 (sky130_fd_sc_hd__o21ai_1)

2 0.01 0.06 0.09 3.22 v core/U215/Y (sky130_fd_sc_hd__o21ai_1)

0.06 0.00 3.22 v core/U96/A1 (sky130_fd_sc_hd__a21o_1)

1 0.00 0.04 0.15 3.37 v core/U96/X (sky130_fd_sc_hd__a21o_1)

0.04 0.00 3.37 v core/U942/CIN (sky130_fd_sc_hd__fa_1)

2 0.01 0.08 0.30 3.67 v core/U942/COUT (sky130_fd_sc_hd__fa_1)

0.08 0.00 3.67 v core/U94/A1 (sky130_fd_sc_hd__a21o_1)

1 0.00 0.04 0.16 3.83 v core/U94/X (sky130_fd_sc_hd__a21o_1)

0.04 0.00 3.83 v core/U905/CIN (sky130_fd_sc_hd__fa_1)

1 0.00 0.07 0.29 4.12 v core/U905/COUT (sky130_fd_sc_hd__fa_1)

0.07 0.00 4.12 v core/U887/CIN (sky130_fd_sc_hd__fa_1)

1 0.00 0.07 0.30 4.42 v core/U887/COUT (sky130_fd_sc_hd__fa_1)

0.07 0.00 4.42 v core/U1370/CIN (sky130_fd_sc_hd__fa_1)

1 0.00 0.07 0.30 4.72 v core/U1370/COUT (sky130_fd_sc_hd__fa_1)

0.07 0.00 4.72 v core/U36/CIN (sky130_fd_sc_hd__fa_2)

2 0.01 0.06 0.29 5.01 v core/U36/COUT (sky130_fd_sc_hd__fa_2)

0.06 0.00 5.01 v core/U87/A (sky130_fd_sc_hd__clkinv_1)

1 0.00 0.03 0.05 5.06 ^ core/U87/Y (sky130_fd_sc_hd__clkinv_1)

0.03 0.00 5.06 ^ core/U613/A2 (sky130_fd_sc_hd__o21ai_1)

1 0.00 0.05 0.04 5.10 v core/U613/Y (sky130_fd_sc_hd__o21ai_1)

0.05 0.00 5.10 v core/U352/CIN (sky130_fd_sc_hd__fa_1)

1 0.00 0.07 0.29 5.39 v core/U352/COUT (sky130_fd_sc_hd__fa_1)

0.07 0.00 5.39 v core/U348/CIN (sky130_fd_sc_hd__fa_1)

1 0.00 0.07 0.30 5.69 v core/U348/COUT (sky130_fd_sc_hd__fa_1)

0.07 0.00 5.69 v core/U35/CIN (sky130_fd_sc_hd__fa_1)

1 0.00 0.07 0.30 5.99 v core/U35/COUT (sky130_fd_sc_hd__fa_1)

0.07 0.00 5.99 v core/U33/CIN (sky130_fd_sc_hd__fa_1)

```

2  0.01  0.08  0.31  6.30 v core/U33/COUT (sky130_fd_sc_hd__fa_1)
    0.08  0.00  6.30 v core/U81/A1 (sky130_fd_sc_hd__a21o_1)
1  0.00  0.04  0.16  6.46 v core/U81/X (sky130_fd_sc_hd__a21o_1)
    0.04  0.00  6.46 v core/U347/CIN (sky130_fd_sc_hd__fa_1)
1  0.00  0.07  0.29  6.75 v core/U347/COUT (sky130_fd_sc_hd__fa_1)
    0.07  0.00  6.75 v core/U351/CIN (sky130_fd_sc_hd__fa_1)
1  0.00  0.07  0.30  7.05 v core/U351/COUT (sky130_fd_sc_hd__fa_1)
    0.07  0.00  7.05 v core/U617/B (sky130_fd_sc_hd__xor2_1)
2  0.01  0.19  0.19  7.24 ^ core/U617/X (sky130_fd_sc_hd__xor2_1)
    0.19  0.00  7.24 ^ core/U618/A (sky130_fd_sc_hd__nand2_2)
15 0.03  0.18  0.14  7.38 v core/U618/Y (sky130_fd_sc_hd__nand2_2)
    0.18  0.00  7.38 v core/U635/B2 (sky130_fd_sc_hd__o22ai_1)
1  0.00  0.14  0.14  7.52 ^ core/U635/Y (sky130_fd_sc_hd__o22ai_1)
    0.14  0.00  7.52 ^ core/CPU_Xreg_value_a4_reg[16][31]/D (sky130_fd_sc_hd__dfxtp_1)
    7.52  data arrival time

```

```

0.00 11.00 11.00 clock clk (rise edge)
    0.00 11.00 clock network delay (ideal)
    0.00 11.00 clock reconvergence pessimism
    11.00 ^ core/CPU_Xreg_value_a4_reg[16][31]/CLK (sky130_fd_sc_hd__dfxtp_1)
-0.07 10.93 library setup time
    10.93 data required time

```

```

-----
    10.93 data required time
    -7.52 data arrival time
-----

```

```

    3.41 slack (MET)

```

----- WNS -----

wns max 0.00

----- TNS -----

tns max 0.00

=====

END OF LIBRARY: sky130_fd_sc_hd__ff_n40C_1v65.lib

=====

=====

LIBRARY: sky130_fd_sc_hd__ff_n40C_1v76.lib

=====

Warning 1140:
/data/Desktop/work/tools/openlane_working_dir/OpenSTA/vsdbabysoc/lib/sky130_fd_sc_hd__ff_n
40C_1v76.lib line 1, library sky130_fd_sc_hd__ff_n40C_1v76 already exists.

Warning 1140:
/data/Desktop/work/tools/openlane_working_dir/OpenSTA/vsdbabysoc/lib/avsddac.lib line 1, library
avsddac already exists.

Warning 1140:
/data/Desktop/work/tools/openlane_working_dir/OpenSTA/vsdbabysoc/lib/avsddac.lib line 1, library
avsddac already exists.

Warning 1140:
/data/Desktop/work/tools/openlane_working_dir/OpenSTA/vsdbabysoc/lib/avsdpll.lib line 1, library
avsdpll already exists.

Warning 1140:
/data/Desktop/work/tools/openlane_working_dir/OpenSTA/vsdbabysoc/lib/avsdpll.lib line 1, library
avsdpll already exists.

----- REPORT CHECKS -----

Warning 168: run_final.tcl line 1, unknown field nets.

Startpoint: core/CPU_imm_a2_reg[10]

(rising edge-triggered flip-flop clocked by clk)

Endpoint: core/CPU_imm_a3_reg[10]

(rising edge-triggered flip-flop clocked by clk)

Path Group: clk

Path Type: min

Fanout	Cap	Slew	Delay	Time	Description
--------	-----	------	-------	------	-------------

	0.00	0.00	0.00		clock clk (rise edge)
		0.00	0.00		clock network delay (ideal)
	0.00	0.00	0.00		^ core/CPU_imm_a2_reg[10]/CLK (sky130_fd_sc_hd__dfxtp_1)
1	0.00	0.03	0.20	0.20	^ core/CPU_imm_a2_reg[10]/Q (sky130_fd_sc_hd__dfxtp_1)
	0.03	0.00	0.20		^ core/CPU_imm_a3_reg[10]/D (sky130_fd_sc_hd__dfxtp_1)
			0.20		data arrival time
	0.00	0.00	0.00		clock clk (rise edge)
		0.00	0.00		clock network delay (ideal)
		0.00	0.00		clock reconvergence pessimism
			0.00		^ core/CPU_imm_a3_reg[10]/CLK (sky130_fd_sc_hd__dfxtp_1)
	-0.02	-0.02			library hold time
			-0.02		data required time

-0.02 data required time

-0.20 data arrival time

0.22 slack (MET)

Startpoint: core/CPU_is_addi_a3_reg

(rising edge-triggered flip-flop clocked by clk)

Endpoint: core/CPU_Xreg_value_a4_reg[1][31]

(rising edge-triggered flip-flop clocked by clk)

Path Group: clk

Path Type: max

Fanout	Cap	Slew	Delay	Time	Description
--------	-----	------	-------	------	-------------

0.00	0.00	0.00	clock clk (rise edge)		
------	------	------	-----------------------	--	--

0.00	0.00	clock network delay (ideal)		
------	------	-----------------------------	--	--

0.00	0.00	0.00	^ core/CPU_is_addi_a3_reg/CLK (sky130_fd_sc_hd__dfxtp_1)		
------	------	------	--	--	--

2	0.01	0.03	0.21	0.21	v core/CPU_is_addi_a3_reg/Q (sky130_fd_sc_hd__dfxtp_1)
---	------	------	------	------	--

0.03	0.00	0.21	v core/U474/A (sky130_fd_sc_hd__nor2_1)		
------	------	------	---	--	--

2	0.00	0.09	0.10	0.31	^ core/U474/Y (sky130_fd_sc_hd__nor2_1)
---	------	------	------	------	---

0.09	0.00	0.31	^ core/U476/A (sky130_fd_sc_hd__nand2_1)		
------	------	------	--	--	--

2	0.01	0.05	0.06	0.37	v core/U476/Y (sky130_fd_sc_hd__nand2_1)
---	------	------	------	------	--

0.05	0.00	0.37	v core/U9/A (sky130_fd_sc_hd__inv_2)		
------	------	------	--------------------------------------	--	--

34	0.14	0.58	0.43	0.81	^ core/U9/Y (sky130_fd_sc_hd__inv_2)
----	------	------	------	------	--------------------------------------

0.58	0.00	0.81	^ core/U479/B (sky130_fd_sc_hd__xor2_1)		
------	------	------	---	--	--

2 0.01 0.13 0.09 0.90 v core/U479/X (sky130_fd_sc_hd__xor2_1)
 0.13 0.00 0.90 v core/U569/A2 (sky130_fd_sc_hd__a21oi_1)
 2 0.01 0.16 0.20 1.10 ^ core/U569/Y (sky130_fd_sc_hd__a21oi_1)
 0.16 0.00 1.10 ^ core/U571/A2 (sky130_fd_sc_hd__o21ai_1)
 2 0.01 0.06 0.07 1.17 v core/U571/Y (sky130_fd_sc_hd__o21ai_1)
 0.06 0.00 1.17 v core/U574/A1 (sky130_fd_sc_hd__a21oi_1)
 2 0.01 0.16 0.16 1.33 ^ core/U574/Y (sky130_fd_sc_hd__a21oi_1)
 0.16 0.00 1.33 ^ core/U576/A2 (sky130_fd_sc_hd__o21ai_1)
 2 0.01 0.06 0.07 1.41 v core/U576/Y (sky130_fd_sc_hd__o21ai_1)
 0.06 0.00 1.41 v core/U581/A1 (sky130_fd_sc_hd__a21oi_1)
 2 0.01 0.16 0.16 1.56 ^ core/U581/Y (sky130_fd_sc_hd__a21oi_1)
 0.16 0.00 1.56 ^ core/U583/A2 (sky130_fd_sc_hd__o21ai_1)
 2 0.01 0.06 0.07 1.64 v core/U583/Y (sky130_fd_sc_hd__o21ai_1)
 0.06 0.00 1.64 v core/U587/A1 (sky130_fd_sc_hd__a21oi_1)
 2 0.01 0.16 0.16 1.80 ^ core/U587/Y (sky130_fd_sc_hd__a21oi_1)
 0.16 0.00 1.80 ^ core/U589/A2 (sky130_fd_sc_hd__o21ai_1)
 2 0.01 0.06 0.07 1.87 v core/U589/Y (sky130_fd_sc_hd__o21ai_1)
 0.06 0.00 1.87 v core/U344/A1 (sky130_fd_sc_hd__a21oi_1)
 2 0.01 0.16 0.16 2.03 ^ core/U344/Y (sky130_fd_sc_hd__a21oi_1)
 0.16 0.00 2.03 ^ core/U213/A2 (sky130_fd_sc_hd__o21ai_1)
 2 0.01 0.06 0.07 2.10 v core/U213/Y (sky130_fd_sc_hd__o21ai_1)
 0.06 0.00 2.10 v core/U343/A1 (sky130_fd_sc_hd__a21oi_1)
 2 0.01 0.16 0.16 2.26 ^ core/U343/Y (sky130_fd_sc_hd__a21oi_1)
 0.16 0.00 2.26 ^ core/U210/A2 (sky130_fd_sc_hd__o21ai_1)
 2 0.01 0.06 0.07 2.33 v core/U210/Y (sky130_fd_sc_hd__o21ai_1)
 0.06 0.00 2.33 v core/U342/A1 (sky130_fd_sc_hd__a21oi_1)
 2 0.01 0.16 0.16 2.49 ^ core/U342/Y (sky130_fd_sc_hd__a21oi_1)

0.16 0.00 2.49 ^ core/U209/A2 (sky130_fd_sc_hd__o21ai_1)

2 0.01 0.06 0.07 2.56 v core/U209/Y (sky130_fd_sc_hd__o21ai_1)

0.06 0.00 2.56 v core/U341/A1 (sky130_fd_sc_hd__a21oi_1)

2 0.01 0.16 0.16 2.72 ^ core/U341/Y (sky130_fd_sc_hd__a21oi_1)

0.16 0.00 2.72 ^ core/U215/A2 (sky130_fd_sc_hd__o21ai_1)

2 0.01 0.06 0.07 2.80 v core/U215/Y (sky130_fd_sc_hd__o21ai_1)

0.06 0.00 2.80 v core/U96/A1 (sky130_fd_sc_hd__a21o_1)

1 0.00 0.03 0.13 2.93 v core/U96/X (sky130_fd_sc_hd__a21o_1)

0.03 0.00 2.93 v core/U942/CIN (sky130_fd_sc_hd__fa_1)

2 0.01 0.07 0.26 3.19 v core/U942/COUT (sky130_fd_sc_hd__fa_1)

0.07 0.00 3.19 v core/U94/A1 (sky130_fd_sc_hd__a21o_1)

1 0.00 0.03 0.14 3.33 v core/U94/X (sky130_fd_sc_hd__a21o_1)

0.03 0.00 3.33 v core/U905/CIN (sky130_fd_sc_hd__fa_1)

1 0.00 0.06 0.25 3.58 v core/U905/COUT (sky130_fd_sc_hd__fa_1)

0.06 0.00 3.58 v core/U887/CIN (sky130_fd_sc_hd__fa_1)

1 0.00 0.06 0.26 3.83 v core/U887/COUT (sky130_fd_sc_hd__fa_1)

0.06 0.00 3.83 v core/U1370/CIN (sky130_fd_sc_hd__fa_1)

1 0.00 0.06 0.26 4.09 v core/U1370/COUT (sky130_fd_sc_hd__fa_1)

0.06 0.00 4.09 v core/U36/CIN (sky130_fd_sc_hd__fa_2)

2 0.01 0.06 0.25 4.34 v core/U36/COUT (sky130_fd_sc_hd__fa_2)

0.06 0.00 4.34 v core/U87/A (sky130_fd_sc_hd__clkinv_1)

1 0.00 0.03 0.04 4.38 ^ core/U87/Y (sky130_fd_sc_hd__clkinv_1)

0.03 0.00 4.38 ^ core/U613/A2 (sky130_fd_sc_hd__o21ai_1)

1 0.00 0.05 0.04 4.42 v core/U613/Y (sky130_fd_sc_hd__o21ai_1)

0.05 0.00 4.42 v core/U352/CIN (sky130_fd_sc_hd__fa_1)

1 0.00 0.06 0.25 4.68 v core/U352/COUT (sky130_fd_sc_hd__fa_1)

0.06 0.00 4.68 v core/U348/CIN (sky130_fd_sc_hd__fa_1)

```

1  0.00  0.06  0.26  4.93 v core/U348/COUT (sky130_fd_sc_hd__fa_1)
    0.06  0.00  4.93 v core/U35/CIN (sky130_fd_sc_hd__fa_1)
1  0.00  0.06  0.26  5.19 v core/U35/COUT (sky130_fd_sc_hd__fa_1)
    0.06  0.00  5.19 v core/U33/CIN (sky130_fd_sc_hd__fa_1)
2  0.01  0.07  0.27  5.46 v core/U33/COUT (sky130_fd_sc_hd__fa_1)
    0.07  0.00  5.46 v core/U81/A1 (sky130_fd_sc_hd__a21o_1)
1  0.00  0.03  0.14  5.60 v core/U81/X (sky130_fd_sc_hd__a21o_1)
    0.03  0.00  5.60 v core/U347/CIN (sky130_fd_sc_hd__fa_1)
1  0.00  0.06  0.25  5.85 v core/U347/COUT (sky130_fd_sc_hd__fa_1)
    0.06  0.00  5.85 v core/U351/CIN (sky130_fd_sc_hd__fa_1)
1  0.00  0.06  0.25  6.10 v core/U351/COUT (sky130_fd_sc_hd__fa_1)
    0.06  0.00  6.10 v core/U617/B (sky130_fd_sc_hd__xor2_1)
2  0.01  0.17  0.17  6.27 ^ core/U617/X (sky130_fd_sc_hd__xor2_1)
    0.17  0.00  6.27 ^ core/U618/A (sky130_fd_sc_hd__nand2_2)
15 0.03  0.17  0.12  6.39 v core/U618/Y (sky130_fd_sc_hd__nand2_2)
    0.17  0.00  6.39 v core/U643/B2 (sky130_fd_sc_hd__o22ai_1)
1  0.00  0.13  0.12  6.52 ^ core/U643/Y (sky130_fd_sc_hd__o22ai_1)
    0.13  0.00  6.52 ^ core/CPU_Xreg_value_a4_reg[1][31]/D (sky130_fd_sc_hd__dfxtp_1)
    6.52  data arrival time

0.00 11.00 11.00 clock clk (rise edge)
    0.00 11.00 clock network delay (ideal)
    0.00 11.00 clock reconvergence pessimism
    11.00 ^ core/CPU_Xreg_value_a4_reg[1][31]/CLK (sky130_fd_sc_hd__dfxtp_1)
-0.05 10.95 library setup time
    10.95 data required time

```

10.95 data required time

-6.52 data arrival time

4.43 slack (MET)

----- WNS -----

wns max 0.00

----- TNS -----

tns max 0.00

=====

END OF LIBRARY: sky130_fd_sc_hd__ff_n40C_1v76.lib

=====

=====

LIBRARY: sky130_fd_sc_hd__ff_100C_1v65.lib

=====

Warning 1140:
/data/Desktop/work/tools/openlane_working_dir/OpenSTA/vsdbabysoc/lib/sky130_fd_sc_hd__ff_100C_1v65.lib line 1, library sky130_fd_sc_hd__ff_100C_1v65 already exists.

Warning 1140:
/data/Desktop/work/tools/openlane_working_dir/OpenSTA/vsdbabysoc/lib/avsdac.lib line 1, library avsdac already exists.

Warning 1140:
/data/Desktop/work/tools/openlane_working_dir/OpenSTA/vsdbabysoc/lib/avsdac.lib line 1, library
avsdac already exists.

Warning 1140:
/data/Desktop/work/tools/openlane_working_dir/OpenSTA/vsdbabysoc/lib/avsdpll.lib line 1, library
avsdpll already exists.

Warning 1140:
/data/Desktop/work/tools/openlane_working_dir/OpenSTA/vsdbabysoc/lib/avsdpll.lib line 1, library
avsdpll already exists.

----- REPORT CHECKS -----

Warning 168: run_final.tcl line 1, unknown field nets.

Startpoint: core/CPU_imm_a2_reg[10]

(rising edge-triggered flip-flop clocked by clk)

Endpoint: core/CPU_imm_a3_reg[10]

(rising edge-triggered flip-flop clocked by clk)

Path Group: clk

Path Type: min

Fanout	Cap	Slew	Delay	Time	Description
--------	-----	------	-------	------	-------------

0.00	0.00	0.00	clock clk (rise edge)		
------	------	------	-----------------------	--	--

0.00	0.00	clock network delay (ideal)		
------	------	-----------------------------	--	--

0.00	0.00	0.00	^ core/CPU_imm_a2_reg[10]/CLK (sky130_fd_sc_hd__dfxtp_1)		
------	------	------	--	--	--

1	0.00	0.03	0.22	0.22	^ core/CPU_imm_a2_reg[10]/Q (sky130_fd_sc_hd__dfxtp_1)
---	------	------	------	------	--

0.03	0.00	0.22	^ core/CPU_imm_a3_reg[10]/D (sky130_fd_sc_hd__dfxtp_1)		
------	------	------	--	--	--

0.22	data arrival time				
------	-------------------	--	--	--	--

0.00	0.00	0.00	clock clk (rise edge)		
------	------	------	-----------------------	--	--

0.00	0.00	clock network delay (ideal)		
------	------	-----------------------------	--	--

0.00	0.00	clock reconvergence pessimism
0.00	^ core/CPU_imm_a3_reg[10]/CLK (sky130_fd_sc_hd__dfxtp_1)	
-0.03	-0.03	library hold time
-0.03	data required time	

-0.03	data required time	
-0.22	data arrival time	

0.25	slack (MET)	

Startpoint: core/CPU_is_addi_a3_reg
(rising edge-triggered flip-flop clocked by clk)
Endpoint: core/CPU_Xreg_value_a4_reg[16][30]
(rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max

Fanout	Cap	Slew	Delay	Time	Description

0.00	0.00	0.00	clock clk (rise edge)		
0.00	0.00	clock network delay (ideal)			
0.00	0.00	0.00	^ core/CPU_is_addi_a3_reg/CLK (sky130_fd_sc_hd__dfxtp_1)		
2	0.01	0.04	0.24	0.24	v core/CPU_is_addi_a3_reg/Q (sky130_fd_sc_hd__dfxtp_1)
	0.04	0.00	0.24	v core/U474/A (sky130_fd_sc_hd__nor2_1)	
2	0.00	0.09	0.09	0.34	^ core/U474/Y (sky130_fd_sc_hd__nor2_1)
	0.09	0.00	0.34	^ core/U476/A (sky130_fd_sc_hd__nand2_1)	

2 0.01 0.07 0.08 0.42 v core/U476/Y (sky130_fd_sc_hd__nand2_1)
 0.07 0.00 0.42 v core/U9/A (sky130_fd_sc_hd__inv_2)
 34 0.15 0.56 0.42 0.83 ^ core/U9/Y (sky130_fd_sc_hd__inv_2)
 0.56 0.00 0.83 ^ core/U479/B (sky130_fd_sc_hd__xor2_1)
 2 0.01 0.16 0.13 0.96 v core/U479/X (sky130_fd_sc_hd__xor2_1)
 0.16 0.00 0.96 v core/U569/A2 (sky130_fd_sc_hd__a21oi_1)
 2 0.01 0.16 0.19 1.16 ^ core/U569/Y (sky130_fd_sc_hd__a21oi_1)
 0.16 0.00 1.16 ^ core/U571/A2 (sky130_fd_sc_hd__o21ai_1)
 2 0.01 0.07 0.09 1.25 v core/U571/Y (sky130_fd_sc_hd__o21ai_1)
 0.07 0.00 1.25 v core/U574/A1 (sky130_fd_sc_hd__a21oi_1)
 2 0.01 0.16 0.15 1.40 ^ core/U574/Y (sky130_fd_sc_hd__a21oi_1)
 0.16 0.00 1.40 ^ core/U576/A2 (sky130_fd_sc_hd__o21ai_1)
 2 0.01 0.07 0.09 1.49 v core/U576/Y (sky130_fd_sc_hd__o21ai_1)
 0.07 0.00 1.49 v core/U581/A1 (sky130_fd_sc_hd__a21oi_1)
 2 0.01 0.16 0.15 1.64 ^ core/U581/Y (sky130_fd_sc_hd__a21oi_1)
 0.16 0.00 1.64 ^ core/U583/A2 (sky130_fd_sc_hd__o21ai_1)
 2 0.01 0.07 0.09 1.73 v core/U583/Y (sky130_fd_sc_hd__o21ai_1)
 0.07 0.00 1.73 v core/U587/A1 (sky130_fd_sc_hd__a21oi_1)
 2 0.01 0.16 0.15 1.89 ^ core/U587/Y (sky130_fd_sc_hd__a21oi_1)
 0.16 0.00 1.89 ^ core/U589/A2 (sky130_fd_sc_hd__o21ai_1)
 2 0.01 0.07 0.09 1.98 v core/U589/Y (sky130_fd_sc_hd__o21ai_1)
 0.07 0.00 1.98 v core/U344/A1 (sky130_fd_sc_hd__a21oi_1)
 2 0.01 0.16 0.15 2.13 ^ core/U344/Y (sky130_fd_sc_hd__a21oi_1)
 0.16 0.00 2.13 ^ core/U213/A2 (sky130_fd_sc_hd__o21ai_1)
 2 0.01 0.07 0.09 2.22 v core/U213/Y (sky130_fd_sc_hd__o21ai_1)
 0.07 0.00 2.22 v core/U343/A1 (sky130_fd_sc_hd__a21oi_1)
 2 0.01 0.16 0.15 2.37 ^ core/U343/Y (sky130_fd_sc_hd__a21oi_1)

0.16 0.00 2.37 ^ core/U210/A2 (sky130_fd_sc_hd__o21ai_1)

2 0.01 0.07 0.09 2.46 v core/U210/Y (sky130_fd_sc_hd__o21ai_1)

0.07 0.00 2.46 v core/U342/A1 (sky130_fd_sc_hd__a21oi_1)

2 0.01 0.16 0.15 2.61 ^ core/U342/Y (sky130_fd_sc_hd__a21oi_1)

0.16 0.00 2.61 ^ core/U209/A2 (sky130_fd_sc_hd__o21ai_1)

2 0.01 0.07 0.09 2.71 v core/U209/Y (sky130_fd_sc_hd__o21ai_1)

0.07 0.00 2.71 v core/U341/A1 (sky130_fd_sc_hd__a21oi_1)

2 0.01 0.16 0.15 2.86 ^ core/U341/Y (sky130_fd_sc_hd__a21oi_1)

0.16 0.00 2.86 ^ core/U215/A2 (sky130_fd_sc_hd__o21ai_1)

2 0.01 0.07 0.09 2.95 v core/U215/Y (sky130_fd_sc_hd__o21ai_1)

0.07 0.00 2.95 v core/U96/A1 (sky130_fd_sc_hd__a21o_1)

1 0.00 0.04 0.14 3.09 v core/U96/X (sky130_fd_sc_hd__a21o_1)

0.04 0.00 3.09 v core/U942/CIN (sky130_fd_sc_hd__fa_1)

2 0.01 0.09 0.28 3.36 v core/U942/COUT (sky130_fd_sc_hd__fa_1)

0.09 0.00 3.36 v core/U94/A1 (sky130_fd_sc_hd__a21o_1)

1 0.00 0.04 0.14 3.50 v core/U94/X (sky130_fd_sc_hd__a21o_1)

0.04 0.00 3.50 v core/U905/CIN (sky130_fd_sc_hd__fa_1)

1 0.00 0.07 0.26 3.76 v core/U905/COUT (sky130_fd_sc_hd__fa_1)

0.07 0.00 3.76 v core/U887/CIN (sky130_fd_sc_hd__fa_1)

1 0.00 0.07 0.27 4.03 v core/U887/COUT (sky130_fd_sc_hd__fa_1)

0.07 0.00 4.03 v core/U1370/CIN (sky130_fd_sc_hd__fa_1)

1 0.01 0.08 0.27 4.31 v core/U1370/COUT (sky130_fd_sc_hd__fa_1)

0.08 0.00 4.31 v core/U36/CIN (sky130_fd_sc_hd__fa_2)

2 0.01 0.07 0.28 4.59 v core/U36/COUT (sky130_fd_sc_hd__fa_2)

0.07 0.00 4.59 v core/U87/A (sky130_fd_sc_hd__clkinv_1)

1 0.00 0.03 0.04 4.62 ^ core/U87/Y (sky130_fd_sc_hd__clkinv_1)

0.03 0.00 4.62 ^ core/U613/A2 (sky130_fd_sc_hd__o21ai_1)

```

1  0.00  0.06  0.05  4.68 v core/U613/Y (sky130_fd_sc_hd__o21ai_1)
    0.06  0.00  4.68 v core/U352/CIN (sky130_fd_sc_hd__fa_1)
1  0.00  0.07  0.26  4.94 v core/U352/COUT (sky130_fd_sc_hd__fa_1)
    0.07  0.00  4.94 v core/U348/CIN (sky130_fd_sc_hd__fa_1)
1  0.00  0.07  0.27  5.21 v core/U348/COUT (sky130_fd_sc_hd__fa_1)
    0.07  0.00  5.21 v core/U35/CIN (sky130_fd_sc_hd__fa_1)
1  0.00  0.07  0.27  5.48 v core/U35/COUT (sky130_fd_sc_hd__fa_1)
    0.07  0.00  5.48 v core/U33/CIN (sky130_fd_sc_hd__fa_1)
2  0.01  0.09  0.29  5.77 v core/U33/COUT (sky130_fd_sc_hd__fa_1)
    0.09  0.00  5.77 v core/U81/A1 (sky130_fd_sc_hd__a21o_1)
1  0.00  0.04  0.14  5.91 v core/U81/X (sky130_fd_sc_hd__a21o_1)
    0.04  0.00  5.91 v core/U347/CIN (sky130_fd_sc_hd__fa_1)
1  0.00  0.07  0.26  6.17 v core/U347/COUT (sky130_fd_sc_hd__fa_1)
    0.07  0.00  6.17 v core/U351/CIN (sky130_fd_sc_hd__fa_1)
2  0.01  0.07  0.38  6.55 ^ core/U351/SUM (sky130_fd_sc_hd__fa_1)
    0.07  0.00  6.55 ^ core/U3/A (sky130_fd_sc_hd__nand2_1)
15 0.03  0.29  0.20  6.75 v core/U3/Y (sky130_fd_sc_hd__nand2_1)
    0.29  0.00  6.75 v core/U696/B2 (sky130_fd_sc_hd__o22ai_1)
1  0.00  0.14  0.13  6.88 ^ core/U696/Y (sky130_fd_sc_hd__o22ai_1)
    0.14  0.00  6.88 ^ core/CPU_Xreg_value_a4_reg[16][30]/D (sky130_fd_sc_hd__dfxtp_1)
    6.88  data arrival time

0.00 11.00 11.00 clock clk (rise edge)
    0.00 11.00 clock network delay (ideal)
    0.00 11.00 clock reconvergence pessimism
    11.00 ^ core/CPU_Xreg_value_a4_reg[16][30]/CLK (sky130_fd_sc_hd__dfxtp_1)
-0.07 10.93 library setup time

```

10.93 data required time

10.93 data required time

-6.88 data arrival time

4.05 slack (MET)

----- WNS -----

wns max 0.00

----- TNS -----

tns max 0.00

=====
END OF LIBRARY: sky130_fd_sc_hd__ff_100C_1v65.lib
=====

=====
LIBRARY: sky130_fd_sc_hd__ff_100C_1v95.lib
=====

Warning 1140:
/data/Desktop/work/tools/openlane_working_dir/OpenSTA/vsdbabysoc/lib/sky130_fd_sc_hd__ff_1
00C_1v95.lib line 1, library sky130_fd_sc_hd__ff_100C_1v95 already exists.

Warning 1140:
/data/Desktop/work/tools/openlane_working_dir/OpenSTA/vsdbabysoc/lib/avsddac.lib line 1, library
avsddac already exists.

Warning 1140:
/data/Desktop/work/tools/openlane_working_dir/OpenSTA/vsdbabysoc/lib/avsddac.lib line 1, library
avsddac already exists.

Warning 1140:
/data/Desktop/work/tools/openlane_working_dir/OpenSTA/vsdbabysoc/lib/avsdpll.lib line 1, library
avsdpll already exists.

Warning 1140:
/data/Desktop/work/tools/openlane_working_dir/OpenSTA/vsdbabysoc/lib/avsdpll.lib line 1, library
avsdpll already exists.

----- REPORT CHECKS -----

Warning 168: run_final.tcl line 1, unknown field nets.

Startpoint: core/CPU_imm_a2_reg[10]

(rising edge-triggered flip-flop clocked by clk)

Endpoint: core/CPU_imm_a3_reg[10]

(rising edge-triggered flip-flop clocked by clk)

Path Group: clk

Path Type: min

Fanout	Cap	Slew	Delay	Time	Description
--------	-----	------	-------	------	-------------

0.00	0.00	0.00	0.00		clock clk (rise edge)
------	------	------	------	--	-----------------------

0.00	0.00				clock network delay (ideal)
------	------	--	--	--	-----------------------------

0.00	0.00	0.00	0.00		^ core/CPU_imm_a2_reg[10]/CLK (sky130_fd_sc_hd__dfxtp_1)
------	------	------	------	--	--

1	0.00	0.03	0.18	0.18	^ core/CPU_imm_a2_reg[10]/Q (sky130_fd_sc_hd__dfxtp_1)
---	------	------	------	------	--

0.03	0.00	0.18	0.18		^ core/CPU_imm_a3_reg[10]/D (sky130_fd_sc_hd__dfxtp_1)
------	------	------	------	--	--

0.18					data arrival time
------	--	--	--	--	-------------------

	0.00	0.00	0.00		clock clk (rise edge)
		0.00	0.00		clock network delay (ideal)
		0.00	0.00		clock reconvergence pessimism
			0.00	^	core/CPU_imm_a3_reg[10]/CLK (sky130_fd_sc_hd__dfxtp_1)
	-0.02	-0.02			library hold time
		-0.02			data required time

		-0.02			data required time
		-0.18			data arrival time

		0.20			slack (MET)
Startpoint: core/CPU_is_addi_a3_reg					
(rising edge-triggered flip-flop clocked by clk)					
Endpoint: core/CPU_Xreg_value_a4_reg[16][30]					
(rising edge-triggered flip-flop clocked by clk)					
Path Group: clk					
Path Type: max					
Fanout	Cap	Slew	Delay	Time	Description

	0.00	0.00	0.00		clock clk (rise edge)
		0.00	0.00		clock network delay (ideal)
	0.00	0.00	0.00	^	core/CPU_is_addi_a3_reg/CLK (sky130_fd_sc_hd__dfxtp_1)
2	0.01	0.03	0.20	0.20	v core/CPU_is_addi_a3_reg/Q (sky130_fd_sc_hd__dfxtp_1)
	0.03	0.00	0.20	v	core/U474/A (sky130_fd_sc_hd__nor2_1)

2 0.00 0.07 0.07 0.27 ^ core/U474/Y (sky130_fd_sc_hd__nor2_1)
 0.07 0.00 0.27 ^ core/U476/A (sky130_fd_sc_hd__nand2_1)
 2 0.01 0.06 0.07 0.33 v core/U476/Y (sky130_fd_sc_hd__nand2_1)
 0.06 0.00 0.33 v core/U9/A (sky130_fd_sc_hd__inv_2)
 34 0.15 0.47 0.35 0.68 ^ core/U9/Y (sky130_fd_sc_hd__inv_2)
 0.47 0.00 0.68 ^ core/U479/B (sky130_fd_sc_hd__xor2_1)
 2 0.01 0.14 0.09 0.77 v core/U479/X (sky130_fd_sc_hd__xor2_1)
 0.14 0.00 0.77 v core/U569/A2 (sky130_fd_sc_hd__a21oi_1)
 2 0.01 0.14 0.15 0.92 ^ core/U569/Y (sky130_fd_sc_hd__a21oi_1)
 0.14 0.00 0.92 ^ core/U571/A2 (sky130_fd_sc_hd__o21ai_1)
 2 0.01 0.07 0.07 0.99 v core/U571/Y (sky130_fd_sc_hd__o21ai_1)
 0.07 0.00 0.99 v core/U574/A1 (sky130_fd_sc_hd__a21oi_1)
 2 0.01 0.14 0.12 1.11 ^ core/U574/Y (sky130_fd_sc_hd__a21oi_1)
 0.14 0.00 1.11 ^ core/U576/A2 (sky130_fd_sc_hd__o21ai_1)
 2 0.01 0.07 0.07 1.18 v core/U576/Y (sky130_fd_sc_hd__o21ai_1)
 0.07 0.00 1.18 v core/U581/A1 (sky130_fd_sc_hd__a21oi_1)
 2 0.01 0.14 0.12 1.30 ^ core/U581/Y (sky130_fd_sc_hd__a21oi_1)
 0.14 0.00 1.30 ^ core/U583/A2 (sky130_fd_sc_hd__o21ai_1)
 2 0.01 0.07 0.07 1.36 v core/U583/Y (sky130_fd_sc_hd__o21ai_1)
 0.07 0.00 1.36 v core/U587/A1 (sky130_fd_sc_hd__a21oi_1)
 2 0.01 0.14 0.12 1.49 ^ core/U587/Y (sky130_fd_sc_hd__a21oi_1)
 0.14 0.00 1.49 ^ core/U589/A2 (sky130_fd_sc_hd__o21ai_1)
 2 0.01 0.07 0.07 1.55 v core/U589/Y (sky130_fd_sc_hd__o21ai_1)
 0.07 0.00 1.55 v core/U344/A1 (sky130_fd_sc_hd__a21oi_1)
 2 0.01 0.14 0.12 1.67 ^ core/U344/Y (sky130_fd_sc_hd__a21oi_1)
 0.14 0.00 1.67 ^ core/U213/A2 (sky130_fd_sc_hd__o21ai_1)
 2 0.01 0.07 0.07 1.74 v core/U213/Y (sky130_fd_sc_hd__o21ai_1)

0.07 0.00 1.74 v core/U343/A1 (sky130_fd_sc_hd__a21oi_1)

2 0.01 0.14 0.12 1.86 ^ core/U343/Y (sky130_fd_sc_hd__a21oi_1)

0.14 0.00 1.86 ^ core/U210/A2 (sky130_fd_sc_hd__o21ai_1)

2 0.01 0.07 0.07 1.93 v core/U210/Y (sky130_fd_sc_hd__o21ai_1)

0.07 0.00 1.93 v core/U342/A1 (sky130_fd_sc_hd__a21oi_1)

2 0.01 0.14 0.12 2.05 ^ core/U342/Y (sky130_fd_sc_hd__a21oi_1)

0.14 0.00 2.05 ^ core/U209/A2 (sky130_fd_sc_hd__o21ai_1)

2 0.01 0.07 0.07 2.12 v core/U209/Y (sky130_fd_sc_hd__o21ai_1)

0.07 0.00 2.12 v core/U341/A1 (sky130_fd_sc_hd__a21oi_1)

2 0.01 0.14 0.12 2.24 ^ core/U341/Y (sky130_fd_sc_hd__a21oi_1)

0.14 0.00 2.24 ^ core/U215/A2 (sky130_fd_sc_hd__o21ai_1)

2 0.01 0.07 0.07 2.31 v core/U215/Y (sky130_fd_sc_hd__o21ai_1)

0.07 0.00 2.31 v core/U96/A1 (sky130_fd_sc_hd__a21o_1)

1 0.00 0.03 0.11 2.42 v core/U96/X (sky130_fd_sc_hd__a21o_1)

0.03 0.00 2.42 v core/U942/CIN (sky130_fd_sc_hd__fa_1)

2 0.01 0.08 0.22 2.63 v core/U942/COUT (sky130_fd_sc_hd__fa_1)

0.08 0.00 2.63 v core/U94/A1 (sky130_fd_sc_hd__a21o_1)

1 0.00 0.03 0.11 2.74 v core/U94/X (sky130_fd_sc_hd__a21o_1)

0.03 0.00 2.74 v core/U905/CIN (sky130_fd_sc_hd__fa_1)

1 0.00 0.06 0.20 2.95 v core/U905/COUT (sky130_fd_sc_hd__fa_1)

0.06 0.00 2.95 v core/U887/CIN (sky130_fd_sc_hd__fa_1)

1 0.00 0.06 0.21 3.16 v core/U887/COUT (sky130_fd_sc_hd__fa_1)

0.06 0.00 3.16 v core/U1370/CIN (sky130_fd_sc_hd__fa_1)

1 0.01 0.07 0.21 3.37 v core/U1370/COUT (sky130_fd_sc_hd__fa_1)

0.07 0.00 3.37 v core/U36/CIN (sky130_fd_sc_hd__fa_2)

2 0.01 0.06 0.22 3.58 v core/U36/COUT (sky130_fd_sc_hd__fa_2)

0.06 0.00 3.58 v core/U87/A (sky130_fd_sc_hd__clkinv_1)

```

1  0.00  0.03  0.03  3.61 ^ core/U87/Y (sky130_fd_sc_hd__clkinv_1)
    0.03  0.00  3.61 ^ core/U613/A2 (sky130_fd_sc_hd__o21ai_1)
1  0.00  0.05  0.04  3.66 v core/U613/Y (sky130_fd_sc_hd__o21ai_1)
    0.05  0.00  3.66 v core/U352/CIN (sky130_fd_sc_hd__fa_1)
1  0.00  0.06  0.21  3.86 v core/U352/COUT (sky130_fd_sc_hd__fa_1)
    0.06  0.00  3.86 v core/U348/CIN (sky130_fd_sc_hd__fa_1)
1  0.00  0.06  0.21  4.07 v core/U348/COUT (sky130_fd_sc_hd__fa_1)
    0.06  0.00  4.07 v core/U35/CIN (sky130_fd_sc_hd__fa_1)
1  0.00  0.06  0.21  4.28 v core/U35/COUT (sky130_fd_sc_hd__fa_1)
    0.06  0.00  4.28 v core/U33/CIN (sky130_fd_sc_hd__fa_1)
2  0.01  0.08  0.22  4.51 v core/U33/COUT (sky130_fd_sc_hd__fa_1)
    0.08  0.00  4.51 v core/U81/A1 (sky130_fd_sc_hd__a21o_1)
1  0.00  0.03  0.11  4.62 v core/U81/X (sky130_fd_sc_hd__a21o_1)
    0.03  0.00  4.62 v core/U347/CIN (sky130_fd_sc_hd__fa_1)
1  0.00  0.06  0.20  4.82 v core/U347/COUT (sky130_fd_sc_hd__fa_1)
    0.06  0.00  4.82 v core/U351/CIN (sky130_fd_sc_hd__fa_1)
2  0.01  0.06  0.30  5.12 ^ core/U351/SUM (sky130_fd_sc_hd__fa_1)
    0.06  0.00  5.12 ^ core/U3/A (sky130_fd_sc_hd__nand2_1)
15 0.04  0.25  0.17  5.28 v core/U3/Y (sky130_fd_sc_hd__nand2_1)
    0.25  0.00  5.28 v core/U696/B2 (sky130_fd_sc_hd__o22ai_1)
1  0.00  0.12  0.11  5.39 ^ core/U696/Y (sky130_fd_sc_hd__o22ai_1)
    0.12  0.00  5.39 ^ core/CPU_Xreg_value_a4_reg[16][30]/D (sky130_fd_sc_hd__dfxtp_1)
    5.39 data arrival time

0.00 11.00 11.00 clock clk (rise edge)
    0.00 11.00 clock network delay (ideal)
    0.00 11.00 clock reconvergence pessimism

```

11.00 ^ core/CPU_Xreg_value_a4_reg[16][30]/CLK (sky130_fd_sc_hd__dfxtp_1)

-0.04 10.96 library setup time

10.96 data required time

10.96 data required time

-5.39 data arrival time

5.57 slack (MET)

----- WNS -----

wns max 0.00

----- TNS -----

tns max 0.00

=====

END OF LIBRARY: sky130_fd_sc_hd__ff_100C_1v95.lib

=====

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ALL CORNER LIBRARIES COMPLETED

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%